



Review

Geographical variation of arsenic distribution in paddy soil, rice and rice-based products: A meta-analytic approach and implications to human health



Supriya Majumder, Pabitra Banik*

Agricultural and Ecological Research Unit, Indian Statistical Institute, 203, B.T. Road, Kolkata, 700108, India

ARTICLE INFO

Keywords:

Arsenic
Rice
Speciation
Meta-analysis
Baby rice
Health risk

ABSTRACT

Arsenic is considered as ubiquitous toxic element belonging to the highest health hazard category. Wide ranges of natural as well as anthropogenic activities are subject to create global arsenic distribution in the broad sense. Rice is the major staple food consumed by world's population on the maximum scale. Growing environment of rice typically attributed by geographical origin may influence on arsenic bioavailability in rice grain. Over exploitation of arsenic contaminated, groundwater resources have recognised as major concern in agricultural perspective for rice production. On the other hand, biogeochemical weathering of arsenic bearing rocks as the geogenic origin, mining activities and application arsenical pesticides are recognised to be well known factors responsible to increase the soil arsenic level. Transfer of arsenic into rice is rightly acquainted from these possible sources of contamination in different regions around the world. Consequently, such substantial geographical variation reflects bioavailability as well as speciation of arsenic in rice. In this manuscript, we discuss the contribution of different arsenic entering pathways in soil-rice systems from regional variability. Furthermore, we attempted to apply the meta-analysis in order to predict the comparative risk assessment on distribution pattern of total and inorganic arsenic in rice commercialised from various rice producing regions of Asia, Europe and US by considering a selected number of data set an extensive range of market basket and field survey. In addition, we finally focus on health risk assessment associated by the consumption of rice and rice-based infant products as the dietary intake from the different of origin. Furthermore, we must detect and categorize the possible source of contamination, which may critically enhance the bioavailability of arsenic in rice in order to minimize the risk. These are the major aspects reviewed here.

1. Introduction

Arsenic is ubiquitous naturally-occurring toxic metalloid found in terrestrial and aquatic environment. In the health concern, it is classified as Group 1 carcinogenic to humans according to International Agency for Research on Cancer (IARC, 2004). Human exposure to elevated levels of arsenic contaminated drinking water possesses the greatest threat to the public health. Uptake of arsenic through drinking water is the toxic route of exposure in human. Adverse effect on human health promotes the worldwide attention where arsenic concentration exceeds the permissible limit of 10 µg per litre (World Health Organization) in drinking water. In agricultural perspective, cultivation of food crops extensively irrigated with arsenic contaminated groundwater are responsible for the toxicity in both soil and crop environment. Thus in another way, food chain transport of arsenic also possesses a potential threat to the human (Meharg et al., 2009; Zhao et al., 2010). Food crops especially rice (*Oryza sativa* L.) is a typical arsenic

accumulating plant (Lin et al., 2015b; Su et al., 2010; Williams et al., 2007a). Based on the picture of rice production, China leads globally followed by India, Indonesia, Bangladesh and other countries (FAO, 2017). Similarly, these countries have taken upper position in per-capita consumption of rice among South and South-East (SE) Asia where rice is the major staple food in comparison to the countries of Europe and United States (US) and parts of the South America.

Occurrence of arsenic in rice, in consequence to geographical variation as well as soil based transfer has been going to be focused globally in recent years across the different countryside in the world. Of course, the proper reason of this variation is not clearly understandable but expected to occurrence of natural as well as anthropogenic activities. From the background of natural source, weathering of arsenic bearing deposited rocks/sediments in alluvial and deltaic origin of SE Asian part can lead to potential mobilization of arsenic in soil. Gangetic delta, including parts of India (West Bengal) and Bangladesh possesses significant arsenic contamination in groundwater resource among SE

* Corresponding author.

E-mail address: banikpabitra@gmail.com (P. Banik).<https://doi.org/10.1016/j.jenvman.2018.12.034>

Received 12 September 2018; Received in revised form 29 November 2018; Accepted 11 December 2018

Available online 20 December 2018

0301-4797/ © 2018 Published by Elsevier Ltd.

Asia due to the presence of polluted aquifers. This has been reflected in soil resulting from arsenic-rich irrigation water application through a number of studies (Kumar et al., 2016b; Srivastava and Sharma, 2013). Arsenic transport pathway over the surface via contaminant irrigation water is threatening to soil as well as plant health. Soil act as the major sink for deposition of arsenic (Sanyal and Nasar, 2002). Irrigation water with elevated arsenic levels leads to high accumulation of arsenic in paddy field of Bangladesh and West Bengal over time. Research shown that application of 1.5 m of irrigation water with arsenic concentration of 0.13 mg L^{-1} could be increased at the rate of $1 \text{ mg kg}^{-1} \text{ crop}^{-1}$ (Duxbury, 2007). Arsenic deposited as $83,000 \mu\text{g kg}^{-1}$ in the paddy field of Bangladesh through groundwater irrigation (Ullah, 1998). Higher level of arsenic was also reported in groundwater as well as soil from China, Taiwan, Vietnam and Thailand (Nordstrom, 2002). Xie and Huang (1998) reported arsenic concentration in Chinese soil of $68,000 \mu\text{g kg}^{-1}$, which enhance the grain arsenic level up to $0.7 \mu\text{g g}^{-1}$. Indeed, higher arsenic concentration as well as distribution in agricultural soil and its uptake by rice through arsenic contaminated irrigation water has been reported by several workers through field studies (Abedin et al., 2002). Cultivation of dry-season rice crop (*boro*) is strongly depended upon the large amount of irrigation water, which is commonly extracted from the shallow tube well.

From different diverse activities, mining, use of arsenical pesticides possesses a serious impact on arsenic load in rice as a point source of pollution (Hite, 2013; Lee, 2006; Liao et al., 2005; Liu et al., 2005; Williams et al., 2007a; Zavala and Duxbury, 2008; Zavala et al., 2008). In the mining impacted regions, arsenic is commonly found at elevated levels causing toxicity in human and ecosystems. Downstream movement of arsenic due to action of heavy rainfall close to mining areas causes the accumulation arsenic in soil and water bodies. Numerous researches have conducted to investigate arsenic concentration in paddy soil as well as rice grain in mining environment in order to assess the potential health risk. On the other hand, organic arsenic pesticides may have the influence on bioavailability as well as toxicity of arsenic in soil through methylation and volatilization (Bednar et al., 2002; Feng et al., 2005). Substantial accumulation of arsenic has been reported in North US soil through the usage of arsenic based pesticide (Sarkar et al., 2005). Ori et al. (1993) found soil arsenic concentration varied over the range of $2800\text{--}73,000 \mu\text{g kg}^{-1}$ with the mean level of $23,200 \mu\text{g kg}^{-1}$ in Louisianan agricultural soil.

Due to the existence of anaerobic condition in paddy soil, rice can accumulate higher levels of arsenic into grain among other cereal crops (Su et al., 2010; Williams et al., 2007b). A number of published literature has well documented the regional variability of arsenic concentration in rice grain. Emerging evidence shows that elevated arsenic content in rice grain under irrigation water environment in Asian countries, especially in India (West Bengal) and Bangladesh has been reflected during the prolonged period of studies (Abedin et al., 2002; Islam et al., 2004; Jahiruddin et al., 2004; Khan et al., 2009; Meharg and Rahman, 2003; Williams et al., 2006). Such association of significant arsenic concentration in rice grain underutilization of irrigation water with higher arsenic levels has been established during *boro* rice cultivation in these regions in particular. Enhanced efficiency of grain arsenic transfer was reported in Bangladeshi rice compared to rice grown in US/Europe with the higher ratio (Adomako et al., 2009). On the other hand, mining and metal processing industries presumably contribute enrichment of arsenic in rice grain. This kind of pathway of arsenic contamination has published in a number of literature for Chinese rice (Liu et al., 2005; Liao et al., 2005; Ma et al., 2016; Williams et al., 2009; Zhu et al., 2008). In EU scenario, rice derived from Italy and Iberian Peninsula region satisfies major proportion of the total production. Arsenic concentration in groundwater above $50 \mu\text{g L}^{-1}$ has been found in several countries of European region with higher bioavailability of arsenic in soil (Cubadda et al., 2010; García-Sánchez et al., 2010; García-Sánchez and Alvarez-Ayuso, 2003; García-Sánchez et al., 2005; Moyano et al., 2009). Wide range of arsenic concentration in

commercialised rice samples has been reported with regional variability from Italian origin (Sommella et al., 2013). Similarly, arsenic in paddy soil of Iberian Peninsula region was found to be geogenic or mining derived (Ramos-Miras et al., 2014). Moreover, scientists have found the usage of arsenic based pesticides as a source of bioavailability of arsenic in US grown rice (Williams et al., 2005, 2007a). This possible source of arsenic contamination in US paddy soil was come under the awareness in early 20th centuries (Reed and Sturgis, 1936). Arsenical pesticide used traditionally in cotton cultivation was strictly banned by the initiative of US EPA in 1987 (Loebenstein, 1994).

Variation in inorganic arsenic content in rice has been reported through a wide range of market basket surveys between the countries (Meharg et al., 2009; Williams et al., 2005). Furthermore, significant variation has been found within the countries in the number of studies (Ma et al., 2016; Sommella et al., 2013; Tenni et al., 2017). Market basket surveys on Asian rice showed that rice grains are mainly dominated by inorganic arsenic where the contribution of dimethylarsinic acid (DMA) is only 20% of total arsenic (Williams et al., 2005; Zhao et al., 2013). Conversely, the majority of rice grown in Europe and US are found to be organic rather than inorganic with wide variability in arsenic speciation (Meharg et al., 2009; Williams et al., 2005; Zhao et al., 2013). Geographical variation primarily governs the relative bioavailability of inorganic as well as organic arsenic in rice from different origin (Zhao et al., 2013). Several authors suggest that rice is the source of dietary exposure of inorganic arsenic (Meliker et al., 2006; Tsuji et al., 2007). However, the risk posed by arsenic exposure from rice is governed by inorganic arsenic concentration in rice grain, and arsenic ingested through grain (Kile et al., 2007; Meharg et al., 2009; Ohno et al., 2007). Generally, it has been found that, inorganic arsenic is more toxic than pentavalent methylated arsenic species (Schoof et al., 1999). A number of studies are available, which calculate the risk assessment for dietary exposure to inorganic arsenic in Bangladesh, China, India, Spain, US, etc. from the consumption of rice and rice-based products.

Meta-analysis is the statistical tool, introduced during the year of 1904. Modern scientific process deals with systematic understanding of the problem through widely synthesized results and to specify the sources of variation in order to explain the outcomes with more precise estimates. The objective of the meta-analysis is the systematic assessment of previous research studies of peer-reviewed articles in the quantitative manner to derive the conclusion through statistical interpretation of results.

There is a need to define global quantification of arsenic concentration in rice grain relative to marginal level from the perspective of risk assessment. In this article, we attempted to apply the meta-analysis approach in order to predict the comparative risk assessment on distribution pattern of total and inorganic arsenic in rice commercialised from different rice producing regions of Asia, Europe and US by considering a selected number of data set including the extensive range of market basket and field survey. Prior studies focused to define the variation of arsenic concentration by the consequence of rice genotype. Variation of methylated arsenic species in rice by environmental influence as well as microbial community has also reported. In this article, we aimed to identify the possible source of geographical variation prevailing on the variability of arsenic in soil and in rice as well. These fundamentally include: geogenic activities, application of contaminated groundwater irrigation, mining and usage of pesticides. There is substantial geographic variation of arsenic concentration in rice grain as well as speciation. Additionally, we also discuss the risk assessment from dietary intake of arsenic by considering the relevance of consumption from rice and rice-based infant products along the different countryside.

2. Environmental chemistry of arsenic in soil

Geographical variation of arsenic in rice is critically attributed by its

chemical behaviour in the environmental component like soil. Arsenic is an element of group 15 in the periodic table mainly exist in four different oxidation states: 3, 0, +3 and +5. In groundwater and soil, the major inorganic arsenic constituents are oxy-anions of arsenite (As^{III}) and arsenate (As^{V}). However, the speciation of these arsenic compounds in different oxidation states is mainly governed by pH, redox potential status and the types of sorbing components (Bissen and Frimmel, 2003; Haque and Johannesson, 2006; Mandal and Suzuki, 2002). Consequently, the redox transformation and adsorption reaction primarily control the mobility, solubility, bioavailability as well as toxicity of arsenic in the soil-water system (Farooqi et al., 2007; Hirata et al., 2006). Thermodynamically, the occurrence of most stable arsenic species, including inorganic arsenious acid (H_3AsO_3) and oxy-anions of arsenate (H_2AsO_4^- and HASO_4^{2-}) found over the pH ranging from 4 to 8 in soil solution (Smith, 2007). H_2AsO_4^- and HASO_4^{2-} are the species of inorganic arsenic acid H_3AsO_4 . Some of the researchers have well explained the stability of arsenic species in soil solution depending upon the Eh-pH diagram through extensive studies. Higher Eh values designate aerobic oxidative environment showing the existence of arsenate oxyanions in considerable proportions at pH 6–8; whereas under anoxic condition, $\text{H}_3\text{AsO}_3/\text{As}(\text{OH})_3$ predominate over a wide range of pH and are replaced by H_2AsO_3^- and HASO_3^{2-} when pH of the solution exceeds 9.2 and 12 respectively. Based on thermodynamic calculations, it has been suggested that H_2AsO_4^- and HASO_4^{2-} predominates below and above the pH of 6.97 respectively (Sadiq, 1997). In the arsenic affected middle Gangetic plain of Bihar, India, As^{III} was contributed as dominant arsenic species (approximately 73%) over As^{V} in the groundwater samples with mean pH and pe values of 7.5 and -2.1 respectively (Kumar et al., 2016b). This is primarily because As^{III} is more weakly bound to soil minerals as compared to As^{V} , which gets to reduce with releasing arsenic into soil solution, specifically under anoxic condition in paddy soil (Dixit and Hering, 2003; Stroud et al., 2011). Therefore, As^{III} is more toxic and relatively more mobile in contaminated soil and aquatic system than As^{V} .

The primary source of arsenic in groundwater is the release of arsenic from solid phases of aquifers, sediments in which arsenic is bound to especially iron oxides, sulphides and organic matter. In soil, aluminium (Al), iron (Fe) and manganese (Mn) oxyhydroxides are recognised as major binding sites for arsenic adsorption (Smedley and Kinniburgh, 2002). Moreover, phosphate ion (PO_4^{3-}) is the main competing ion for arsenic adsorption, especially As^{V} because of the similarities in terms of dissolution, adsorption and ion exchange between them (Dzombak, 1990). However, several studies have been carried out to determine pH dependent arsenic sorption on soil mineral surfaces. Results revealed that the solubility of As^{V} increased with the increase in pH ranges between 3 and 8 in soil solution whereas solubility of As^{III} decreased with decreasing pH in soil (Jain and Loeppert, 2000; Smith et al., 1999; Tyler and Olsson, 2001). The adsorption of As^{V} is maximum on Fe oxyhydroxides (FeOOH) upon pH around 4; in case of As^{III} , the maximum is found within the pH range of 7–8.5 (Fitz and Wenzel, 2002; Mahimairaja et al., 2005). Furthermore, in most of the cases, the adsorption follows the Langmuir model with $r > 0.7$ maintaining a thermodynamically mediated process (Dousova et al., 2012).

Arsenic behaviour in soil is critically sensitive to Fe redox chemistry (Fitz and Wenzel, 2002; Takahashi et al., 2004). This is because both As^{III} and As^{V} adsorb, mainly to the surface of FeOOH in soil, which is strongly dependent on redox conditions. Smedley and Kinniburgh (2002) reported that the dominance of iron oxide was believed to control arsenic bioavailability and mobility in soil. However, Xu et al. (2008) observed a rapid and marked mobilization of As^{III} in soil solution under flooded condition. Previous studies have shown the possible mechanism responsible for this mobility in soil solution: desorption of As^{III} from the active adsorption surface of FeOOH and the reductive dissolution of FeOOH which actually releases coprecipitated or adsorbed arsenic (Hamon et al., 2004; Masscheleyn et al., 1991; Takahashi et al., 2004). Later, Lafferty and Loeppert (2005) also

reported the influence of Fe oxide mineral on arsenic retention, solubility, and released in submerged soil. On the contrary, under aerobic soil condition, less arsenic mobility is due to the strong adsorption of As^{V} on mineral constituents like Fe and Al oxyhydroxides and aluminosilicates at pH below 8.5 (Goldberg, 2002). Al hydroxide bears the higher adsorption capacity for binding of As^{V} than As^{III} (Manning and Goldberg, 1997). Therefore, the release of arsenic can be occurred in soil containing the higher amount of Al hydroxide when As^{V} is reduced to As^{III} . On the other hand, Manning et al. (1998) and Dixit and Hering (2003) similarly found the higher surface adsorption capacity of FeOOH for As^{III} as compared to As^{V} at pH above 6. In another comparative study, Herbel and Fendorf (2006) reported the much higher rate of desorption of As^{III} from FeOOH than that of As^{V} . These findings can be explained by the possible structural mechanism as surface complex of As^{III} includes both outer sphere complexes (Goldberg and Johnston, 2001) and inner sphere complexes (Stachowicz et al., 2006) whereas As^{V} forms an only inner sphere complex via ligand exchange mechanism (Sherman and Randall, 2003). However, particle size distribution and organic matter content may have the influence on soil arsenic availability (Lombi et al., 2000; Martin et al., 2015; Williams et al., 2011).

3. Factors detecting the source of variability of arsenic in soil

3.1. Geogenic and irrigation water activities

The evaluation of areas vulnerable to arsenic contamination has become worldwide attention during recent years to scientists as well as policy makers. In many of the Asian countries, in particular, elevated arsenic concentration in groundwater as well as soil is known to be influenced by arsenic containing alluvial and deltaic sediments. Biogeochemical weathering of such deposits can lead to release of arsenic into water bodies resulting from the contamination of groundwater (Nordstrom, 2002). Numerous reports suggest that presence of arsenic has been found as reduced condition in the vast tract of alluvial aquifers of Bengal delta with a number of mobilization and fixation reaction (Ahmed et al., 2004; Martin et al., 2014). Geogenic origin of arsenic contamination has been well pictured in this region (Sanz et al., 2007). The occurrence of reduced arsenic is mainly governed by the reductive dissolution of arsenic rich iron oxy hydroxides in dispersed aquifer sediments (Pigna et al., 2015; Yamaguchi et al., 2011). A study also postulated that arsenic bearings aquifers developed from young Holocene sediments are the key indicator of arsenic risk for SE Asian countries (Winkel et al., 2008). In some different approach, Kumar et al. (2016b) observed the risk of arsenic in groundwater was associated with the sediment colour tools, and they also concluded that darkness of the sediment promotes highly reduced state, which could increase the rapid mobilization of arsenic into groundwater. They also reported the association of As^{III} and As^{V} occurs with siderite sediment in the aquifer from middle Gangetic plain of Bihar (India). The nature and property of soil parent materials appear to be the main factor that critically determines arsenic availability in soils. However, arsenic concentration in soil and sediment is associated with a good correlation (Patel et al., 2005).

Arsenic contamination in groundwater of Indo Gangetic Plain and Bangladesh region is the principal practice of discussion during past few decades increasing the risk of agricultural soils through irrigation. Rice field may act as the sink of arsenic where groundwater irrigation is applied and contributes a little beat of recharge to the aquifer via irrigation return-flow through sub soil (Neumann et al., 2011). Bioavailability of arsenic in paddy soil and groundwater irrigation has long been associated by the wide range of research studies. Srivastava and Sharma (2013) estimated elevated arsenic concentration of $5400\text{--}15,430\text{ }\mu\text{g kg}^{-1}$ in Indian soil. In recent study, total arsenic concentration varied from 3528 to $14,690\text{ }\mu\text{g kg}^{-1}$ in agricultural soil of middle Gangetic plain of Bihar (India) (Kumar et al., 2016b). These

estimated values exceed the typical average value of $2000 \mu\text{g kg}^{-1}$ of arsenic in sedimentary and igneous rock (Mandal and Suzuki, 2002). The situation of higher arsenic concentration has recognised in Bangladesh paddy field where extensive groundwater is used for irrigation. Total arsenic concentration in the range of $4000\text{--}137,900 \mu\text{g kg}^{-1}$ was determined in agricultural soil of Bangladesh (Khan et al., 2010b). In another study, arsenic concentration ranged from $880\text{--}4960 \mu\text{g kg}^{-1}$, $3110\text{--}8900 \mu\text{g kg}^{-1}$, $17,600\text{--}65,000 \mu\text{g kg}^{-1}$ in Feni, Dhamrai and Faridpur district of Bangladesh respectively (Ahsan et al., 2009). Meharg and Rahman (2003) reported wide variability of soil arsenic concentration in the range of $3200\text{--}42,500 \mu\text{g kg}^{-1}$ in Bangladeshi paddy field. In the earlier study, Mandal and Suzuki (2002) reported that arsenic concentration ranged from $2000\text{--}4600 \mu\text{g kg}^{-1}$ in soils of South Korea, $6700\text{--}9100 \mu\text{g kg}^{-1}$ in soils from Thailand and $8500\text{--}10,300 \mu\text{g kg}^{-1}$ in Japanese paddy field.

Besides Asian countries, the existence of geogenic arsenic contamination was predicted in the shallow aquifer of Louisiana, US (Yang et al., 2014). Typical arsenic concentration in soil was reported in the part of European countries. In Ambles valley (Spain) region, Ramos-Miras et al. (2014) determined total arsenic concentration ranged from $7300\text{--}35,000 \mu\text{g kg}^{-1}$ and $2200\text{--}6800 \mu\text{g kg}^{-1}$ in the soil developed from granite parent materials and alluvium-colluvium sedimentary materials respectively. Several studies reported wide variability of total arsenic concentration in Spanish soil with the range of $4300\text{--}16,000 \mu\text{g kg}^{-1}$ (De Miguel et al., 2007) and $500\text{--}116,000 \mu\text{g kg}^{-1}$ (Díez et al., 2007). On the other hand, regional variability in soil arsenic was also reported from the major rice producing region of Italy ranging from $6300\text{--}20,000 \mu\text{g kg}^{-1}$ with the mean of $9300 \mu\text{g kg}^{-1}$ (Tenni et al., 2017).

3.2. Seasonal variation

Bioavailability of arsenic in soil may be influenced by seasonal variation. A number of studies have been reported to find seasonal gains and losses of soil arsenic over the extensive range of environmental and agricultural conditions under rice-growing areas from Bangladesh and West Bengal (India). Boro rice cultivation extracts large amount of irrigation water from shallow aquifers. Numerous field studies conducted in Bangladesh indicate much of the arsenic retained in top soil during boro season. Dittmar et al. (2010) estimated annually $4.4 \pm 0.4 \text{ kg ha}^{-1} \text{ a}^{-1}$ arsenic input with irrigation water during boro season. Over three years of periodic study their result depicted that average arsenic of $2.4 \pm 0.4 \text{ kg ha}^{-1} \text{ a}^{-1}$ was deposited during boro season, whether the loss of $2.0 \text{ kg ha}^{-1} \text{ a}^{-1}$ arsenic in average was occurred during the monsoon period. Loss of arsenic from upper soil layer of Bangladeshi paddy field based on mass balance calculation was consistent with the findings during monsoon season (Dittmar et al., 2007; Roberts et al., 2010; Saha and Ali, 2007). However, in other aspects, Khan et al. (2009) conducted an experiment with sequentially grown boro and aman rice (winter rice) cultivars in paddy soil of Bangladesh to determine the effect of added arsenic via irrigation water in undisturbed soil columns during flooded rice culture. Results found that about 70% retention of added arsenic was retained in the zone of maximum root activity and then gradually decreased as the soil depth. In another experiment, Khan et al. (2010a) studied the mobility and vertical distribution of soil arsenic by the consequences of arsenic applied through irrigation water and direct surface soil application over 2 years of the growth cycle under paddy cultivation. They observed that 39.3–47.6% of applied arsenic retained in the surface soil horizon (0–15 cm).

In the Bengal deltaic region of West Bengal (India), Shrivastava et al. (2017) observed gradual increase of arsenic accumulation in agricultural soil layer as $19,400 \mu\text{g kg}^{-1}$, $27,170 \mu\text{g kg}^{-1}$ and $41,240 \mu\text{g kg}^{-1}$ with depth of 0–5 cm, 5–10 cm and 5–15 cm, respectively during monsoon period of study. In another investigation, addition of $120\text{--}420 \text{ mg}$ arsenic per m^2 through application of

$1200\text{--}1400 \text{ mm}$ contaminated irrigation water ha^{-1} has found in summer rice cultivation (Sarkar et al., 2012). The accumulation and availability of arsenic in paddy soil are also influenced by diurnal and seasonal variation in microbiological activity of soil bacteria and algae, in particular (Brammer and Ravenscroft, 2009).

3.3. Mining activities

Abandoned metal mines may contribute significant source of arsenic contamination to the environment. Due to activities of heavy rainfall and strong winds downstream movement of arsenic from the vicinity of mining areas and contamination of natural soil-water systems has been reported (Jung and Thornton, 1997). Large amount of arsenic released into paddy soils through contaminated irrigation water by the activities of mining and smelting is a very common practice. However, the irrigation activities contribute much higher bioavailability of arsenic over mining activities as irrigation water carries the large amount of soluble inorganic arsenic. For instance, in a comparative study, Khan et al. (2010b) found total arsenic concentration in the range of $59,500\text{--}78,600 \mu\text{g kg}^{-1}$ and $6800\text{--}46,100 \mu\text{g kg}^{-1}$ in the mining impacted Chinese and UK soil samples respectively whereas soils of Bangladesh were found to be reached up to $137,900 \mu\text{g kg}^{-1}$ from application of irrigation water. In China, Hunan province has found to be the hot spot area within the country with non-ferrous mining, lead-zinc ore mining and smelting activities (Liao et al., 2005; Ma et al., 2016; Williams et al., 2009). The area is known to be the major rice producing area within the country. Another study reported wide variability of soil arsenic concentration ranged between $3700\text{--}120,300 \mu\text{g kg}^{-1}$ with the mean of $30,400 \mu\text{g kg}^{-1}$ in the emerged Pb-Zn mine affected region of Renhua and Lechang of China (Li et al., 2014). According to the recent investigation conducted by Kwon et al. (2017), a wide variation of arsenic concentration has reported in the range of $2850\text{--}70,400 \mu\text{g kg}^{-1}$ with the mean of $64,400 \mu\text{g kg}^{-1}$ in Korean paddy soil samples influenced by mining activities. Release of arsenic through weathering of arsenopyrite into the headwater stream was reported at the mining site of Virginia (US) (Basu and Schreiber, 2013). Mining impacted arsenic contamination in soil was also evaluated from part of the European countries. In mining derived Peninsula region (Europe), total arsenic content varied from $2300\text{--}17,000 \mu\text{g kg}^{-1}$ with the median concentration of $8700 \mu\text{g kg}^{-1}$ (Signes-Pastor et al., 2016a). Elevated levels of arsenic content were also reported in mining affected soils of Spain with the variation of $17,000\text{--}21,000 \mu\text{g kg}^{-1}$ (Álvarez-Ayuso et al., 2008); $5500\text{--}150,000 \mu\text{g kg}^{-1}$ (García-Sánchez et al., 2010) and $200\text{--}59,000 \mu\text{g kg}^{-1}$ (García-Sánchez and Álvarez-Ayuso, 2003).

4. Arsenic concentration and speciation in commercial rice from variation in geographic origin

4.1. Variability of total arsenic concentration in rice

A number of studies have been reported on rice grain arsenic availability in wide range of field, and market based survey in different geographical region (Table 1). Evidence suggests that arsenic content in rice is likely to be varied with country of origin as well as different production sites within the country. Variability of arsenic concentration in raw rice among the Asian countries shows that the levels are typical compared to other regions of world. Large variability in soil arsenic level from irrigation water is reflected towards elevated grain arsenic from paddy field of Bangladesh and India in particular (Meharg and Rahman, 2003; Roychowdhury, 2008). Preceding rice surveys conducted by Meharg and Rahman (2003) and Duxbury et al. (2003) focussed on rice grown in Rice Research Institute (BRRI) experimental station and specific geographic regions from different districts of Bangladesh with $n = 9$ and $n = 150$, respectively. The level of arsenic in rice samples ranged from $0.043\text{--}0.206 \mu\text{g g}^{-1}$ d. wt. ($n = 9$) and

Table 1The concentration ($\mu\text{g g}^{-1}$) of total arsenic, inorganic arsenic and organic arsenic species in commercial rice grain from different countries.

Country	Total arsenic mean (range), n	Inorganic arsenic mean (range), n	Organic arsenic mean (range)	% of Inorganic arsenic range	Survey range	References
Bangladesh	0.515 (0.058–1.835), 13	–	–	–	Field study (district level)	Meharg and Rahman (2003)
	0.138 (0.043–0.206), 9	–	–	–	Field study (Experimental trial)	Meharg and Rahman (2003)
	(0.01–0.47), 150	–	–	–	Field study	Duxbury et al. (2003)
	0.13 (0.03–0.30), 15	–	–	–	Market basket	Williams et al. (2005)
	0.11–0.36	–	–	–	Market basket	Williams et al. (2006)
	(0.04–0.92), 189-aman	–	–	–		
	0.09–0.51	–	–	–		
	(0.08–0.91), 129- boro	–	–	–		
	0.18–0.31, 12-aman	0.11–0.22, 12	–	60–72		
	0.21–0.27, 9-boro	0.17–0.22, 9	–	81–83		
	(0.57–0.69)	–	–	–	Field study	Rahman et al. (2006)
	(0.60–0.70)	–	–	–	Field study	Rahman et al. (2007)
	0.69 (0.41–0.98)	0.31 (0.23–0.39)	0.10 (0.07–0.12)	55–64		Sun et al. (2008)
	0.143 (0.020–0.557), 214	–	–	–	Field study	Rahman et al. (2009)
	0.13 (0.02–0.33), 144	0.08 (0.01–0.21), 15	–	–	Market basket	Meharg et al. (2009)
India	0.056 (0.038–0.073), 6	0.068, 1	< dl	93	Market basket	Rahman et al. (2014)
	0.126 (0.003–0.680), 965	(0.011–0.502), 162	–	–	District level	Islam et al. (2017)
	0.11 (0.04–0.18), 2	–	–	–	Village level	Roychowdhury (2002)
	0.21 (0.11–0.44), 6	–	–	–	Village level	Roychowdhury (2002)
	0.33 (0.18–0.43), 3	–	–	–	Village level	Roychowdhury (2002)
	0.232, (0.040–0.605), 11	–	–	–	Contaminated field	Roychowdhury et al. (2003)
	0.05 (0.03–0.08), 11	0.03 (0.02–0.05), 11	(< dl-0.01)	36–67	Market basket	Williams et al. (2005)
	0.05 (0.03–0.07), 10	–	–	–	Market basket	Williams et al. (2006)
	0.239 (0.043–0.662), 34	–	–	–	Contaminated field	Roychowdhury (2008)
	0.190 (0.045–0.386), 18	–	–	–	Contaminated field	Roychowdhury (2008)
	0.07 (0.01–0.18), 133	0.03 (0.02–0.07), 12	–	–	Market basket	Meharg et al. (2009)
	0.06–0.58, 68- aman	–	–	–	Contaminated field	Bhattacharya et al. (2010)
	0.05–0.54, 68- boro	–	–	–		
	0.092 (0.077–0.108), 15	0.094, 1	< dl	98	Market basket	Rahman et al. (2014)
	0.283 (0.028–0.961), 29	0.194 (0.025–0.576), 29	0.016 (0.002–0.056)	51–102	Household	Halder et al. (2014)
China	0.14 (0.02–0.46), 124	0.16 (0.07–0.38), 21	–	–	Market basket	Meharg et al. (2009)
	(0.009–0.624), 215	–	–	–	Field study (mining)	Zhu et al. (2008)
	0.22 (0.019–0.586), 19	0.16 (0.079–0.379), 19	0.048 (0.009–0.147)	56.0–92.5	Market basket	Zhu et al. (2008)
	0.26 (0.17–0.39), 15	–	–	–	Field study (mining)	Li et al. (2014)
	0.117 (0.063–0.155), 1653	0.091 (0.048–0.116), 1653	–	–	Field study	Huang et al. (2015)
	0.13 (0.05–0.25), 43	0.119 (0.049–0.211), 43	0.096 (0.01–0.31)	84–97	Field study (mining)	Ma et al. (2016)
	^a 0.057 (0–0.226), 260	0.044 (nd-0.210), 260	0.013 (nd-0.160)	–	Field study (metal polluted)	Lin et al. (2014)
	0.10 (0.06–0.14), 15	–	–	–	Market basket	Williams et al. (2006)
	(0.127–0.175), 3	(0.066–0.114), 3	–	52–65	Market basket	Torres-Escribano et al. (2008)
	0.14 (0.01–0.39), 54	–	–	–	Market basket	Meharg et al. (2009)
Taiwan	0.172 (0.159–0.196), 12	0.168, 2	< dl	86–99	Market basket	Rahman et al. (2014)
	(0.19–0.76), 3	(0.11–0.51), 3	(0.04–0.11)	58–67	Market basket	Williams et al. (2005)
	0.12, 51- white rice	0.07, 51	0.014	50	Market basket	Chen et al. (2016)
	0.22, 13-brown rice	0.11, 13	0.012	58		
Philippines	0.07 (0.00–0.25), 22	–	–	–	Market basket	Williams et al. (2006)
Japan	0.19 (0.07–0.42), 26	–	–	–	Market basket	Meharg et al. (2009)
Korea	0.247 (0.104–0.774), 40	–	–	–	Field study (mining)	Kwon et al. (2017)
Pakistan	0.082 (0.073–0.088), 9	0.083, 1	< dl	96	Market basket	Rahman et al. (2014)
Australia	0.03 (0.02–0.04), 5	–	–	–	Market basket	Williams et al. (2006)
	0.270 (0.188–0.438), 21	(0.165–0.276), 12	(< dl-0.115)	58–94	Market basket	Rahman et al. (2014)
Italy	0.15 (0.07–0.33), 38	0.11 (0.07–0.16), 5	–	–	Market basket	Meharg et al. (2009)
	(0.07–0.47), 101	(0.001–0.20), 101	(nd-0.28)	32–96	Market basket	Sommella et al. (2013)
	0.400 (0.253–0.547), 6	0.271, 1	0.107	49	Market basket	Rahman et al. (2014)
	0.155 (0.049–0.523), 168	0.102 (0.041–0.184), 168	–	31–94	Field study	Tenni et al. (2017)
Spain	0.21 (0.098–0.406), 11-white	0.095 (0.062–0.145), 11	–	33–67	Market basket	Torres-Escribano et al. (2008)
	0.20 (0.145–0.302), 11-brown	0.154 (0.081–0.253), 11	–	55–93		
	0.20 (0.05–0.82), 76	–	–	–	Market basket	Meharg et al. (2009)
	0.28 (< dl-1.16), 30	0.25 (0.15–0.55), 11	0.29 (0.12–0.67), 11	–	Market basket	Tattibayeva et al. (2016)
	^a (0.037–0.433), 113-polished	(0.027–0.175), 113	(0.003–0.333)	40–73	Market basket	Signes-Pastor et al. (2016)
	^a (0.083–0.619), 20- brown	(0.053–0.247), 20	(0.006–0.366)	40–64		
France	0.28 (0.09–0.56), 33	–	–	–	Market basket	Meharg et al. (2009)
Portugal	0.36 (0.22–0.78), 20	0.18 (0.10–0.30), 13	0.23 (0.12–0.70)	–	Market basket	Tattibayeva et al. (2015)

(continued on next page)

Table 1 (continued)

Country	Total arsenic mean (range), n	Inorganic arsenic mean (range), n	Organic arsenic mean (range)	% of Inorganic arsenic range	Survey range	References
US	0.28 (0.17–0.40), 8-white long	0.08 (0.02–0.11, 8	0.16 (0.09–0.26)	10–39	Market basket	Williams et al. (2005)
	0.23 (0.11–0.34), 4-brown long	0.11 (0.06–0.14), 4	0.09 (0.04–0.15)	41–61		
	0.17 (0.10–0.30), 27 (California)	–	–	–	Market basket	Williams et al. (2007)
	0.30 (0.15–0.66), 107 (South central region)	–	–	–		
	0.20 (0.008–0.714), 112	–	–	–	Market basket	Zavala and Duxbury (2008)
	0.29 (0.19–0.71), 24	0.116 (0.072–0.173), 24	0.162 (0.040–0.320)	22–80	Market basket	Zavala et al. (2008)
	0.25 (0.03–0.66), 163	0.10 (0.05–0.15), 10	–	–	Market basket	Meharg et al. (2009)

< dl-below detection limit.

nd-not detected.

^a Σ of arsenic species has considered.

0.010–0.470 $\mu\text{g g}^{-1}$ at 14 percent of moisture content ($n = 150$). Moreover, they also reported the arsenic concentration in the range of 0.058–1.835 $\mu\text{g g}^{-1}$ d. wt. ($n = 13$) in rice grain samples of different varieties collected from the western part of Bangladesh. This is in consistent with the findings of Islam et al. (2004). Based on the seasonal variation in Bangladesh, Williams et al. (2006) reported that arsenic concentration in grain varied from 0.04 to 0.92 $\mu\text{g g}^{-1}$ d. wt. and 0.04–0.91 $\mu\text{g g}^{-1}$ d. wt. for *aman* and *boro* rice, respectively. Rahman et al. (2006) observed an average concentration of 0.57–0.69 $\mu\text{g g}^{-1}$ d. wt. arsenic in raw rice grain from arsenic affected areas of Bangladesh. Significant levels of arsenic in Bangladeshi rice were also reported from field study as well as market basket survey: 0.03–0.30 $\mu\text{g g}^{-1}$ d. wt. (Williams et al., 2005); 0.6–0.7 $\mu\text{g g}^{-1}$ d. wt. (Rahman et al., 2007); 0.41–0.98 $\mu\text{g g}^{-1}$ d. wt. (Sun et al., 2008); 0.02–0.56 $\mu\text{g g}^{-1}$ d. wt. (Rahman et al., 2009); 0.02–0.33 $\mu\text{g g}^{-1}$ d. wt. (Meharg et al., 2009); 0.038–0.073 $\mu\text{g g}^{-1}$ d. wt. (Rahman et al., 2014). Recently, a district level household survey was conducted by Islam et al. (2017) in order to assess the regional distribution of arsenic in rice grain. The variation of total arsenic was estimated from 0.003 to 0.680 $\mu\text{g g}^{-1}$ d. wt. with the mean of 0.126 $\mu\text{g g}^{-1}$ d. wt. ($n = 965$).

In India, Roychowdhury et al. (2003) found wide variability in arsenic content varied from 0.04 to 0.61 $\mu\text{g g}^{-1}$ d. wt. in rice collected from Murshidabad district of West Bengal. Considerable higher amount of arsenic was also observed from his further study in Nadia (0.04–0.39 $\mu\text{g g}^{-1}$ d. wt.) and Murshidabad (0.04–0.66 $\mu\text{g g}^{-1}$ d. wt.) districts of West Bengal showing the existence of arsenic rich iron pyrite in Bengal delta sediments, which is normally responsible for release of arsenic in aquifer (Roychowdhury, 2008). Bhattacharya et al. (2010) observed total grain arsenic accumulation in the range of 0.16–0.58 $\mu\text{g g}^{-1}$ d. wt. under rice cultivation during *aman* and *boro* seasons from different arsenic affected areas of Nadia district, West Bengal. In recent household survey on Nadia district, Halder et al. (2014) reported elevated arsenic concentration in raw rice ranging from 0.025 to 0.961 $\mu\text{g g}^{-1}$ d. wt. with the mean of 0.283 $\mu\text{g g}^{-1}$ d. wt. ($n = 29$). Wide range of arsenic content in Indian rice grain was also reported by various researchers: 0.11–0.44 $\mu\text{g g}^{-1}$ d. wt. (Roychowdhury et al., 2002); 0.03–0.08 $\mu\text{g g}^{-1}$ d. wt. (Williams et al., 2005); 0.077–0.108 $\mu\text{g g}^{-1}$ d. wt. (Rahman et al., 2014).

Besides irrigation water, possible risk in terms of arsenic accumulation in rice grain through mining activities has now made a global issue. Abandoned metal mining as environmental pollutants may lead substantial arsenic accumulation in rice grain. Kwon et al. (2017) found an average arsenic concentration of 0.25 $\mu\text{g g}^{-1}$ d. wt. in polished rice grain collected from abandoned metal mining areas in Korea where the arsenic concentration ranged from 0.104 to 0.774 $\mu\text{g g}^{-1}$ d. wt. Arsenic concentration in the range of 0.18–0.39 $\mu\text{g g}^{-1}$ d. wt. in the Chinese rice from Pb-Zn mining impacted paddy field was reported based on the field survey (Li et al., 2014). In another comparative study, Zhu et al.

(2008) approached to characterize the enrichment of rice grain arsenic level collected from national market as well as mining impacted regions of Hunan and Guangdong in southern China. Total arsenic concentration was varied in the range of 0.074–0.624 $\mu\text{g g}^{-1}$ d. wt. in mining and industrial area in which majority of the samples were estimated to exceed the base line level of Chinese regulatory limit (0.150 $\mu\text{g g}^{-1}$) in rice grain. Compared to the mining scale of survey, average grain arsenic concentration was 0.114 $\mu\text{g g}^{-1}$ d. wt. with the range of 0.080–0.155 $\mu\text{g g}^{-1}$ d. wt. for the rice samples collected from Chinese supermarket. Another recent investigation showed that the variation in total arsenic concentration ranged from 0.050 to 0.253 $\mu\text{g g}^{-1}$ d. wt. with the mean of 0.129 $\pm$ 0.046 $\mu\text{g g}^{-1}$ d. wt. ($n = 43$) in rice grain grown from paddy soil of mining affected Hunan region, China (Ma et al., 2016). In an earlier survey, Meharg et al. (2009) obtained similar trend of arsenic distribution in Chinese polished rice samples varied from 0.02 to 0.46 $\mu\text{g g}^{-1}$ d. wt. with the mean of 0.14 $\mu\text{g g}^{-1}$ d. wt. ($n = 124$).

Levels of arsenic content in rice of few Asian countries were 0.19–0.76 $\mu\text{g g}^{-1}$ d. wt. (Williams et al., 2005), 0.12–0.22 $\mu\text{g g}^{-1}$ d. wt. (Chen et al., 2016) from Taiwan; 0.07–0.42 $\mu\text{g g}^{-1}$ d. wt. from Japan (Meharg et al., 2009); 0.06–0.14 $\mu\text{g g}^{-1}$ d. wt. (Williams et al., 2006), 0.127–0.175 $\mu\text{g g}^{-1}$ d. wt. (Torres-Escribano et al., 2008) from Thailand; 0.00–0.25 $\mu\text{g g}^{-1}$ d. wt. from Philippines (Williams et al., 2006).

In European scenario, the Italian grown rice makes the fulfilment of 50% for the domestic need, and the remaining is exported to other countries, mainly France, Germany, UK (Italian Grain and Feed Report, 2012). Sommella et al. (2013) observed the total arsenic content in Italian rice varied with geographic origin and type. Arsenic content varied from 0.07 to 0.47 $\mu\text{g g}^{-1}$ d. wt. ($n = 101$) in the commercial rice grain samples collected from Italian supermarkets. Tenni et al. (2017) also found significant differences in grain arsenic concentration ranging from 0.049 to 0.523 $\mu\text{g g}^{-1}$ d. wt. ($n = 168$) with rice producing areas in Italy. Meharg et al. (2009) analysed Italian rice ($n = 38$) samples and observed total arsenic content of 0.07–0.33 $\mu\text{g g}^{-1}$ d. wt. which was comparatively lower than rice collected from France (0.09–0.56 $\mu\text{g g}^{-1}$ d. wt., $n = 33$) and Spain (0.05–0.82 $\mu\text{g g}^{-1}$ d. wt., $n = 76$) among the European origin. Significant relationship was obtained between different Italian rice cultivars with the grain arsenic level in the range of 0.12–0.28 $\mu\text{g g}^{-1}$ d. wt. (D'ilio et al., 2002). Wide variability of arsenic content was also reported in Spanish grown commercial rice samples. Ranges of total arsenic concentration were 0.098–0.406 $\mu\text{g g}^{-1}$ d. wt. and 0.145–0.302 $\mu\text{g g}^{-1}$ d. wt. in Spanish white and brown rice, respectively (Torres-Escribano et al., 2008). A recent study conducted by Signes-Pastor et al. (2016a) showed the geographical variation in arsenic content of commercial rice grain collected from the major rice growing regions of Iberian Peninsula across the geographic region of Portugal and Spain in Europe through the market surveys. Total arsenic species content varied from 0.037 to 0.433 $\mu\text{g g}^{-1}$ d. wt. and

0.083–0.619 $\mu\text{g g}^{-1}$ d. wt. in polished and brown rice grain respectively. In another investigation, mean total arsenic content of 0.28 $\mu\text{g g}^{-1}$ d. wt. ($n = 30$) and 0.36 $\mu\text{g g}^{-1}$ d. wt. ($n = 20$) was reported in milled rice from Spain and Portugal respectively (Tattibayeva et al., 2016).

In the view of global comparison, Zavala and Duxbury (2008) categorized different rice producing countries as the level of arsenic content by fixing global normal range between 0.08 and 0.20 $\mu\text{g g}^{-1}$ in grain on a large scale of data set. They found significantly less arsenic content with the mean of 0.051–0.053 $\mu\text{g g}^{-1}$ d. wt. in rice from India and Pakistan than comparatively higher arsenic content with the mean of 0.062–0.089 $\mu\text{g g}^{-1}$ d. wt. from Bangladesh, Venezuela and Thailand. Moreover, they found mean total arsenic concentration of 0.20 mg kg^{-1} in US rice ($n = 112$) with the widest range of 0.008–0.714 $\mu\text{g g}^{-1}$ d. wt. through the extensive range of the survey. This higher trend in US rice was consistent with the findings by Meharg et al. (2009) where arsenic concentration ranged from 0.03 to 0.66 $\mu\text{g g}^{-1}$ d. wt. with the mean value of 0.25 $\mu\text{g g}^{-1}$ d. wt. ($n = 163$). Williams et al. (2007a) also published higher concentration of arsenic (0.15–0.66 $\mu\text{g g}^{-1}$ d. wt.) for rice grown in the South-Central US as compared to rice grown in California (0.10–0.30 $\mu\text{g g}^{-1}$ d. wt.). This result was consistent with the previous findings reported by Heitkemper et al. (2001) in rice from US. The higher bioavailability of arsenic in US grown rice was linked to the application of arsenic containing chemical in agriculture and therefore, soil arsenic contamination occurred. This link was previously suggested by Williams et al. (2005) and Williams et al. (2007a). From another market-based survey Williams et al. (2005) found US long-grain white rice, and brown rice contained mean total arsenic level of 0.28 $\mu\text{g g}^{-1}$ d. wt. ($n = 8$) and 0.23 $\mu\text{g g}^{-1}$ d. wt. ($n = 4$) respectively. In the similar study, they obtained comparatively lower level of arsenic in Bangladeshi and European rice with the mean of 0.13 $\mu\text{g g}^{-1}$ d. wt. ($n = 15$) and 0.18 $\mu\text{g g}^{-1}$ ($n = 7$) d. wt., respectively.

Variation in total arsenic concentration (0.02–0.04 $\mu\text{g g}^{-1}$ d. wt.) was also reported in Australian rice by Williams et al. (2006). In recent study, Rahman et al. (2014) showed that significant variation in arsenic content ranging from 0.188 to 0.438 $\mu\text{g g}^{-1}$ d. wt. ($n = 21$) in Australian grown rice. On an average, Italian rice contributed maximum concentration of grain arsenic level with 0.4 $\mu\text{g g}^{-1}$ d. wt. on sale followed by rice imported from Thailand (0.172 $\mu\text{g g}^{-1}$ d. wt.), India (0.92 $\mu\text{g g}^{-1}$ d. wt.), Pakistan (82 $\mu\text{g g}^{-1}$ d. wt.) and Bangladesh (0.56 $\mu\text{g g}^{-1}$ d. wt.).

4.2. Variability of arsenic speciation in rice

The relative concentration of inorganic and organic arsenic in rice grain seems to depend on rice genotypes and rice-growing environment (Islam et al., 2016; Li et al., 2009; Liu et al., 2006; Xu et al., 2008; Norton et al., 2009) as well as geographic origin (Zhao et al., 2013). Contribution of inorganic and organic arsenic species in rice grain of different origin has been shown in Table 1. Inorganic arsenic was detected as predominant arsenic species for Bangladeshi rice grown in *aman* and *boro* seasons. The mean recovery of inorganic arsenic species was 66 and 82% of total arsenic content during *aman* and *boro* seasons, respectively (Williams et al., 2006). Much higher contribution of inorganic arsenic was estimated in the range of 59–98% over organic arsenic in rice grain collected from different districts of Bangladesh (Islam et al., 2017). The variation of inorganic arsenic concentration in the range of 0.011–0.502 $\mu\text{g g}^{-1}$ d. wt. was reported. Majority of arsenic species was also found to be inorganic contributing in the range of 81.0–96.9% of total arsenic from Indian grown rice collected from the contaminated field (Roychowdhury, 2008). Another recent household survey from West Bengal depicted the wide proportion rate of inorganic arsenic with the variation from 0.025 to 0.576 $\mu\text{g g}^{-1}$ d. wt. in raw rice (Halder et al., 2014). However, irrigation water is likely to be a major component to increase the level of inorganic arsenic in grain during

boro rice cultivation. Recently, Signes-Pastor et al. (2016a) found inorganic arsenic and DMA were the dominant species in commercial rice samples collected from Iberian Peninsula region of the European origin through market survey where inorganic arsenic accounted for 41–97%. Inorganic arsenic and DMA were ranged between 0.027 and 0.175 $\mu\text{g g}^{-1}$ d. wt. and 0.003–0.333 $\mu\text{g g}^{-1}$ d. wt., 0.053–0.247 $\mu\text{g g}^{-1}$ d. wt. and 0.006–0.366 $\mu\text{g g}^{-1}$ d. wt. for polished and brown rice, respectively. The brown rice belonged to higher proportion of inorganic arsenic as compared to polished rice. In another survey, comparatively higher mean inorganic arsenic content of 0.25 $\mu\text{g g}^{-1}$ d. wt. and 0.18 $\mu\text{g g}^{-1}$ d. wt. was reported in Spanish and Portuguese milled rice samples, respectively (Tattibayeva et al., 2016). An earlier investigation from Spain suggested that mean inorganic arsenic concentration of 0.095 $\mu\text{g g}^{-1}$ d. wt. ($n = 11$) and 0.154 $\mu\text{g g}^{-1}$ d. wt. ($n = 11$) in white and brown rice, respectively (Torres-Escribano et al., 2008). In a recent investigation by Tenni et al. (2017), wide variability in proportion of inorganic arsenic content from 31 to 94% was found in polished rice samples grown in different rice producing areas of Italy. The estimated mean inorganic arsenic concentration was $102 \pm 26 \mu\text{g g}^{-1}$ d. wt. ($n = 168$) with the mean recovery of 72.11% of total arsenic. On the other hand, Sommella et al. (2013) found strong influence of geographic origin on concentration of inorganic arsenic in 4 rice growing regions in Italy with wide variability between 0.4 and 96% of total arsenic. Williams et al. (2005) reported that the variability of inorganic arsenic content was 53–65% of market rice collected from Italy. Based on extensive datasets, Meharg et al. (2009) found geographic variability in inorganic arsenic content in white rice across the different countries. Higher mean inorganic arsenic of 0.16 $\mu\text{g g}^{-1}$ d. wt. ($n = 21$) and 0.11 $\mu\text{g g}^{-1}$ d. wt. ($n = 5$) were reported from Chinese and Italian origin respectively as compared to Bangladesh (0.08 $\mu\text{g g}^{-1}$ d. wt., $n = 15$) and India (0.03 $\mu\text{g g}^{-1}$ d. wt., $n = 12$). Variation of inorganic arsenic was reported in the range of 0.05–0.15 $\mu\text{g g}^{-1}$ d. wt. ($n = 10$) from US grown rice. Earlier survey reported variability of inorganic arsenic content ranged from 0.033 to 0.211 $\mu\text{g g}^{-1}$ on the wet weight basis in US grown white rice (Lamont, 2003).

Speciation studies on Chinese rice exhibited extensive variability in inorganic arsenic content. In the mining impacted region of China, Zhu et al. (2008) found wide variability of inorganic arsenic content in unpolished rice grain with the range of 0.078–0.257 $\mu\text{g g}^{-1}$ d. wt. which accounted for 40–93% of total arsenic species. Rice samples despite exhibited higher proportion of inorganic arsenic collected from the market as well as mining field site. The result was consistent with the recent study conducted by Ma et al. (2016) where the mean inorganic arsenic concentration was $0.119 \pm 0.034 \mu\text{g g}^{-1}$ ($n = 43$) in Chinese rice samples of Hunan province. In another mining impacted region of Guangdong province, the inorganic arsenic ranged from not detected to 0.158 $\mu\text{g g}^{-1}$ d. wt. (Lin et al., 2015a). Huang et al. (2015) analysed 1653 Chinese milled rice samples from 11 provinces. They got significant variation in inorganic arsenic concentration among rice producing and consuming regions ranged from 0.048 to 0.116 $\mu\text{g g}^{-1}$ d. wt.

Recently in the rice-growing region of Australia, Rahman et al. (2014) estimated mean inorganic arsenic content of $0.276 \pm 0.025 \mu\text{g g}^{-1}$ d. wt., $0.178 \pm 0.022 \mu\text{g g}^{-1}$ d. wt., $0.165 \pm 0.08 \mu\text{g g}^{-1}$ d. wt., $0.177 \pm 0.013 \mu\text{g g}^{-1}$ d. wt. in organic brown rice, whole brown rice, organic white rice, and sticky shushi rice respectively. The contribution of inorganic arsenic was estimated to be 63, 62, 58 and 94% of total arsenic in 4 different rice types respectively.

Variability in the contribution of DMA in rice grown from different origin is presumably of special significance. Li et al. (2009) and Carey et al. (2010) reported much higher translocation efficiency of DMA from rice root to shoot, and from shoot to grain compared to inorganic arsenic. In contrast, Williams et al. (2005) approached to design the survey of samples from globally rice producing regions like Asia, Europe, US, etc. in order to assess the variation in proportion of inorganic arsenic in commercial polished rice related to dietary exposure.

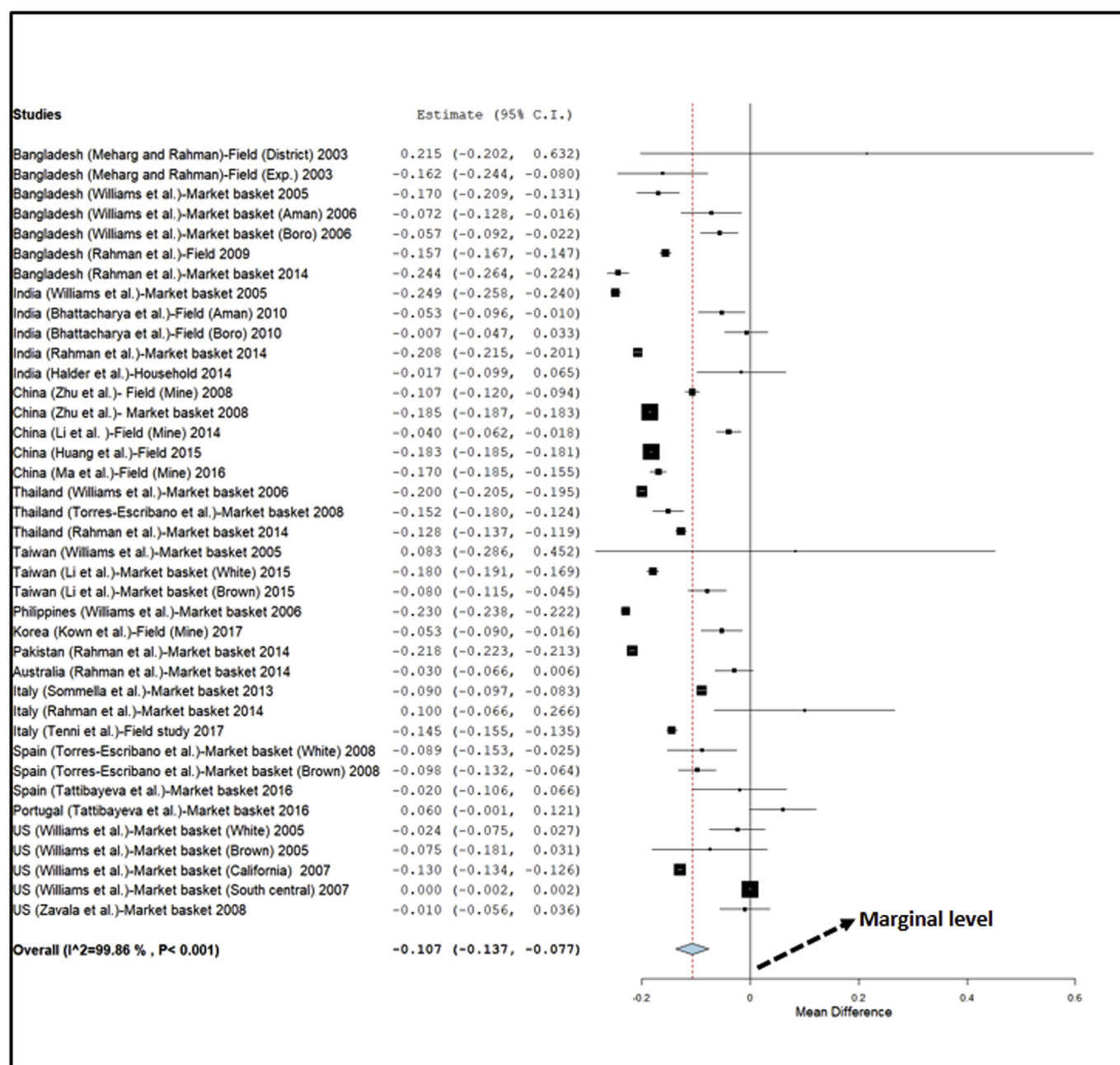


Fig. 1. Forest plot showing the weighted mean difference of total arsenic concentration in rice grain between study level and marginal level with their respective confidence intervals and weight in the meta-analysis together with the heterogeneity statistics.

Around 80% of mean inorganic arsenic was found in rice from Bangladesh and India. They found comparatively lower mean inorganic arsenic content of 64% and 42% in European and US rice respectively. However, the US rice was found to be DMA type in particular. Higher proportion of DMA was also reported in US rice bought in markets in Beijing with the average value of $0.137 \mu\text{g g}^{-1}$ d. wt. (Zhu et al., 2008). As per the global-scale approach, Zavala et al. (2008) categorized rice samples ($n = 105$) from different source of origin into inorganic and DMA type. Majority of US rice ($n = 15$ out of 24) was likely to be found as DMA type in consistent, whereas rice collected from Asian and European origin was predominantly as inorganic type. Recovered DMA was found to be 50–81% of the total arsenic in rice grain from Arkansas, California and Texas. They also produced a strong linear relationship (0.817) with the upward trend of DMA as the total arsenic is increased. Conversely in relationship between inorganic and total arsenic, Meharg et al. (2009) obtained higher slope values for rice produced in

Bangladesh (0.719) and India (0.796) compared to much lower in US rice (0.275). In more recent study, Zhao et al. (2013) reported the relationship between total and inorganic arsenic was hyperbolic in nature of the maximum approaching value of $0.15 \mu\text{g g}^{-1}$ for US rice compared to Asian rice, which possessed a linear relationship of 0.78. The trend of European rice was the intermediate between US and Asia. In similar relationship, variable slope factor in the range of 0.091–0.748 was reported for rice produced across the geographic region of Spain and Portugal (Signes-Pastor et al., 2016a).

5. Meta-analysis-quantitative assessment of pooled data of total and inorganic arsenic in rice from different origin

The objective of the meta-analysis is to find out the relative comparison between the different conducted peer-reviewed studies of specific outcomes. We searched the literature published in the period from

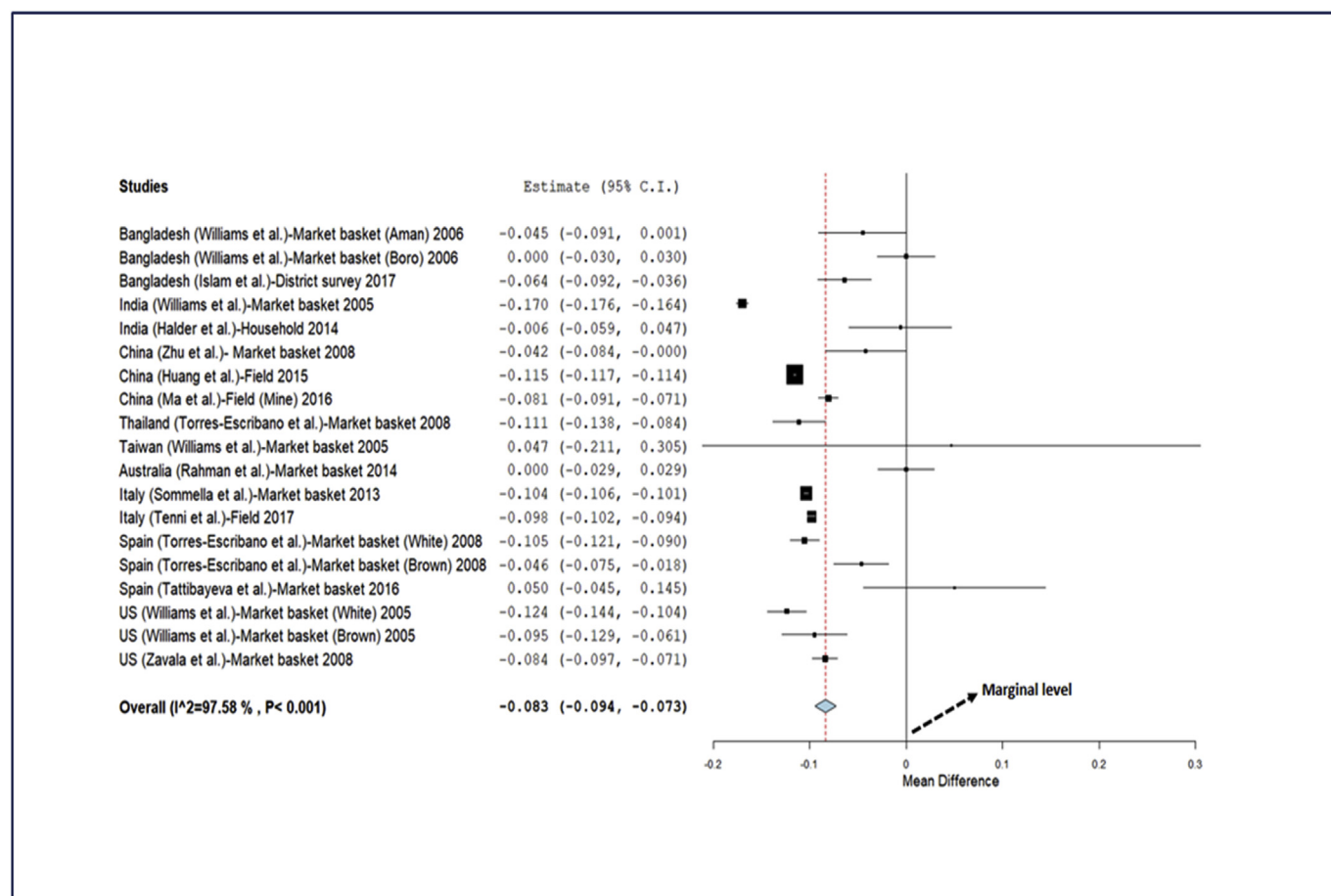


Fig. 2. Forest plot showing the weighted mean difference of inorganic arsenic concentration in rice grain between study level and marginal level with their respective confidence intervals and weight in the meta-analysis together with the heterogeneity statistics.

2002 to 2017 using the source of Pubmed database in order to select the studies according to quality of reporting. The area of major keywords was confined to arsenic in rice grain, arsenic accumulation in rice, inorganic arsenic in rice, arsenic speciation in rice, regional variation of arsenic in rice. From > 700 published articles, we excluded the articles which contained no original data. The studies on solution culture were also rejected in the selection process. We screened the remaining of individual studies from different origin based on the title and abstract of article. From the wide range of inspection on availability of raw dataset, we finally included 40 studies published in the peer-reviewed journals resulting in the impact of regional variability of total and inorganic arsenic in rice either from market basket as well as field level of investigations. To evaluate the comparison of relative risk on arsenic content in rice grain from different countries, we compiled the data set of above individual studies in a quantitative manner through meta-analysis. The sample size and mean values were collected from selected studies. The standard deviation of each observation was evaluated prior to construct the meta-analysis. The effect of size value was reflected by the mean difference between the observed level of arsenic from different countries and marginal level of arsenic in rice grain. The final result was expressed on weighted mean difference as continuous outcomes for statistical parameter at the confidence level of 95% between the group of individual study and marginal level of total and inorganic arsenic in rice (guideline value recommended by WHO) individually. Consequently, the weight of each study was depended on the sample size. Forest plot was created to summarise the information on individual studies in the meta-analysis which also provides the visual suggestion of the amount of heterogeneities. A vertical line in central part of the plot represented the lack of difference between the study

group and marginal level commonly known as no effect or zero effect line. At this point, the mean difference was zero. Mean difference values from individual studies were represented by subsequent squares while the size of the squares indicated the weight of the studies and effect of estimate. The subsequent ends of each horizontal segment showed bias-corrected 95% bootstrap confidence intervals (CI) which were symmetrical about the mean.

To predict the comparative risk estimation of total arsenic in rice from different origin of Asia, Europe and US, 39 numbers of studies were taken into account. Fig. 1 shows the list of input studies with their effect sizes. Each of the individual studies was sequenced in chronological order according to the country of origin. The interpretation of meta-analysis was conducted by concerning the maximum base line marginal value of total arsenic in rice recommended from WHO ($0.3 \mu\text{g g}^{-1}$) (JECFA, 2012). From the continuous random-effects model the overall summary weighted mean value of -0.107 (within the permissible marginal value of $0.3 \mu\text{g g}^{-1}$) showed statistically significant with $p < 0.001$ and inconsistency index of 99.87% indicated substantial heterogeneity. The significance of negative value indicated that the estimated total arsenic concentration was found to be within the safe limit. The relative weights of individual study and summary of statistics were represented in Supplementary. From the resulted mean difference effect size, the majority of the rice grown from Bangladesh, China, Thailand, Italy and Spain were found to be statistically significant at the meta-analysis level with $p < 0.001$ as the subsequent confidence intervals did not overlap the zero effect line. Especially in Chinese rice where mean difference was maintained a safe distance from summary level in maximum of study. This may be due to the maintaining of strict Chinese regulatory limit of arsenic in rice grain

(0.15 $\mu\text{g g}^{-1}$). But the study of Bhattacharya et al. (2010) in *boro* rice and Halder et al. (2014) from Indian rice showed the statistically insignificant estimates which was associated with irrigation water application with elevated arsenic concentration. Consistently similar trend was resulted in the study from Australian rice (Rahman et al., 2014), Spanish rice (Tattibayeva et al., 2016) and majority of US rice. But the rice grown in California was found to be in lower risk (Williams et al., 2007a). The mean difference values in some of the studies from Bangladesh through the district level surveys (Meharg and Rahman, 2003), Taiwan (Williams et al., 2005), Italy (Rahman et al., 2014), and Portugal (Tattibayeva et al., 2016) critically exceeded the zero effect line which may contribute to higher risk of consumption.

In contrast, meta-analysis of inorganic arsenic was conducted with 19 observations assuming the marginal value of 0.2 $\mu\text{g g}^{-1}$ in rice grain (JECFA, 2012). The overall meta-analysed summary effect value of -0.083 which showed statistically significant with $p < 0.001$ and inconsistency index of 97.58% indicates substantial heterogeneity (Fig. 2). The rice produced in Thailand, Italy and Spain contained lower mean value of inorganic arsenic resulting in statistically significant difference from marginal level. In contrast, the confidence intervals overlapped the line of zero effects in case of study from Bangladesh (Williams et al., 2005), India (Halder et al., 2014) showing the non-significant difference need to be required much attention. The utilisation of irrigation water from the shallow tube well in these regions reflected the higher accumulation of inorganic arsenic in rice grain for *boro* cultivation, in particular, which favours the associated risk. On the other hand, the Chinese market rice showed non-significant difference from the study conducted by Zhu et al. (2008) with elevated mean inorganic arsenic concentration of 0.16 $\mu\text{g g}^{-1}$ in rice grain. Other studies from Chinese rice were resulted significant difference from marginal level (Huang et al., 2015; Ma et al., 2016). Rice grown in Australia as reported by Rahman et al. (2014) favoured the risk as the mean difference of inorganic arsenic was situated on zero effect line. In comparison to the total arsenic, all the US rice produced significant difference from marginal level. This could be exhibited the presence of higher percentage of DMA, which encountered the proportion of inorganic arsenic in US rice grain as we previously discussed.

6. Arsenic concentration and speciation in baby rice and rice-based products

Inorganic arsenic content in baby rice used to feed infants may be associated with variation in geographic origin as well as the manufacturer throughout the world (Table 2). Probably lowest total arsenic concentration of 0.30 $\mu\text{g kg}^{-1}$ d. wt. in the pure infant rice was reported from Sweden (Ljung et al., 2011). In the more recent study, Signes-Pastor et al. (2016b) analysed the concentration of arsenic species in baby rice samples ($n = 29$) purchased from UK supermarket. The summation of arsenic species concentration varied from 0.063 to 0.334 $\mu\text{g g}^{-1}$ d. wt. with the wide variation of inorganic arsenic and DMA content ranged from 0.056 to 0.268 $\mu\text{g g}^{-1}$ d. wt. 0.030–0.123 $\mu\text{g g}^{-1}$ d. wt. respectively. Results showed that, inorganic arsenic concentration represented 51.4–84.6% of the total arsenic species. They also found that between 29 samples, milled baby rice samples of organically origin contributed the median percentage of 55.1 for inorganic arsenic level over the non-organic source of 67.1%. Median inorganic arsenic contribution was 70.6% and 80.9% in rice cereals and rice crackers irrespective of production method and manufacturer respectively. In an earlier study, Meharg et al. (2008) determined inorganic arsenic content in baby rice obtained from 3 different supermarkets of Aberdeen, UK. The authors reported concentration of inorganic arsenic varying from 0.06 to 0.16 $\mu\text{g g}^{-1}$ d. wt. ($n = 17$) with the median of 0.11 $\mu\text{g g}^{-1}$ d. wt. The concentration range was lower than the safe level of inorganic arsenic of 0.2 $\mu\text{g g}^{-1}$ d. wt. recommended by FAO/WHO (JECFA, 2012). The variation of inorganic arsenic and DMA ranged from 33.1 to 68.8% and 21.9–54.4%

Table 2
The concentration ($\mu\text{g g}^{-1}$) of total arsenic, inorganic arsenic and organic arsenic species in infant rice and rice-based products manufacturing from different countries.

Country	Product type	Total arsenic Mean (range), n	Inorganic arsenic Mean (range), n	Organic arsenic Mean (range)	Inorganic arsenic % (range)	References
UK	Baby rice	(0.12–0.47), 17	(0.06–0.16), 17	(0.03–0.23)	33.1–66.8	Meharg et al. (2008)
	Pure infant rice	0.237 (0.137–0.394), 5	0.162 (0.107–0.262), 5	0.083 (0.028–0.265)	34–84	Carbonell-Barrachina et al. (2012)
	Baby rice	(0.084–0.334) ^a , 29	(0.056–0.268), 29	(0.024–0.127)	51.4–86.6	Signes-Pastor et al. (2016)
	Rice cereals	(0.042–0.396) ^a , 53	(0.008–0.323), 53	(0.006–0.085)	14.2–89.6	Signes-Pastor et al. (2016)
	Rice crackers	(0.019–0.328) ^a , 97	(0.018–0.211), 97	(0.002–0.178)	44.8–100.0	Signes-Pastor et al. (2016)
Spain	Pure infant rice	0.181 (0.036–0.315), 7	0.085 (0.010–0.111), 7	0.093 (0.025–0.178)	5–75	Carbonell-Barrachina et al. (2012)
	Gluten free infant rice	0.126 (0.046–0.315), 13	0.069 (0.029–0.121), 13	0.056 (0.010–0.198)	36–89	Carbonell-Barrachina et al. (2012)
	Infant cereals with gluten	0.033 (0.015–0.065), 9	0.026 (0.010–0.049), 9	nd-0.005	90–100	Carbonell-Barrachina et al. (2012)
	Gluten free infant rice	0.057 (0.043–0.080), 5	-	-	-	Burló et al. (2012)
	Infant cereals with gluten	0.024 (0.016–0.039), 5	-	-	-	Burló et al. (2012)
US	Rice-based food	0.0–0.285), 11	0.079–0.140, 17	-	-	Munera-Picazo et al. (2014)
	Pure infant rice	0.253 (0.164–0.496), 5	0.125 (0.095–0.159), 5	0.127 (0.035–0.334)	31–81	Carbonell-Barrachina et al. (2012)
	Baby rice	(0.060–0.373), 85	(0.039–0.254), 85	(0.016–0.216)	22–82.9	USFDA (2013)
	Rice cereals	(0.050–0.723), 105	(0.023–0.283), 105	(0.008–0.507)	18.8–87.7	USFDA (2013)
	Rice cracker	(0.009–1.931), 199	(0.008–0.273), 199	(0.002–0.498)	18.4–97.6	USFDA (2013)
China	Pure infant rice	0.135 (0.070–0.353), 14	0.114 (0.052–0.247), 14	0.033 (0.021–0.063)	62–88	Carbonell-Barrachina et al. (2012)

nd-not detected.

^a Σ of arsenic species has considered.

respectively. In another comparative study, Carbonell-Barrachina et al. (2012) determined the variation of total arsenic concentration in the range of $0.036\text{--}0.496\text{ }\mu\text{g g}^{-1}$ fresh wt. in pure infant rice samples ($n = 31$) between 4 different countries. The mean concentration of inorganic arsenic was arranged in the sequence of UK ($0.162 \pm 0.029\text{ }\mu\text{g g}^{-1}$ fresh wt.) > US ($0.125 \pm 0.014\text{ }\mu\text{g g}^{-1}$ fresh wt.) > China ($0.114 \pm 0.015\text{ }\mu\text{g g}^{-1}$ fresh wt.) > Spain ($0.085 \pm 0.010\text{ }\mu\text{g g}^{-1}$ fresh wt.). But the mean contribution of DMA was sequenced in the decreasing order as Spain ($45 \pm 5\%$) > US ($43 \pm 8\%$) > UK ($29 \pm 9\%$) > China ($23 \pm 2\%$). Moreover, the mean ratio of inorganic arsenic to total arsenic was 0.76, 0.71, 0.53 and 0.51 from China, UK, US and Spain respectively. In the same study, they also found the ratio in the range of 0.36–0.89 with the mean of 0.64 in Spanish gluten-free infant rice. The values agreed quite well with the result obtained from Spanish rice-based food products feed to children of celiac disease where the values ranged from 0.48 to 1.11 with the mean of 0.67 ± 0.06 (Munera-Picazo et al., 2014). In another study, Sun et al. (2009) reported relatively higher range of the ratio with the mean of 0.81 in rice products and Japanese rice condiments. Burló et al. (2012), furthermore, estimated the significantly higher arsenic concentration in Spanish gluten-free rice compared to rice cereals using gluten with the mean of $0.057\text{ }\mu\text{g g}^{-1}$ d. wt. and $0.024\text{ }\mu\text{g g}^{-1}$ d. wt. respectively.

A number of authors have focussed to find out relation between the level of arsenic content (total and inorganic) and percentage of the rice content in the samples of rice-based products. Munera-Picazo et al. (2014) obtained a strong positive relationship between the percentages of rice used as ingredients in different manufacturing Spanish food products and the concentration level of total ($R^2 = 0.8909$) and inorganic arsenic ($R^2 = 0.9003$). The content of the rice varied from 5 to 95% with the variation of total and inorganic arsenic concentration in the range of not detected to $0.285\text{ }\mu\text{g g}^{-1}$ fresh wt. and not detected to $0.170\text{ }\mu\text{g g}^{-1}$ fresh wt. respectively. A same positive correlation was also reported by (Carbonell-Barrachina et al., 2012) in Spanish rice-based foods for inorganic arsenic ($R^2 = 0.8504$) and total arsenic ($R^2 = 0.884$).

7. Associated health risk from consumption of arsenic contaminated rice

7.1. Health risk assessment from dietary intake of arsenic through rice

Emerging evidence suggested that consumption of arsenic contaminated rice may have the adverse health effect to human. A number of countries are found to be detected in high risk where rice the principal dietary food. Consequently, dietary intake of arsenic through rice has now become a serious health concern. Species estimation of arsenic seems to assess the health risk associated with its toxicity to human. Fig. 3 is summarized the impact of different arsenic entering pathways on rice grain arsenic bioavailability and human health risk from dietary exposure of arsenic. Reports suggest that the bioavailability of inorganic in rice is in the order of 90% (Juhász et al., 2006). Therefore, rice is likely to be the possible source of exposure of inorganic arsenic to the inhabitants also. To assess the carcinogenicity on potential health risk of arsenic exposure, scientists have introduced the parameters like hazard quotient (HQ) and incremental lifetime cancer risk (CR). The HQ is associated to predict the potentiality of non-carcinogenic health risk while CR is found to be assessed the carcinogenic risk. As per the 72nd meeting of Joint Expert Committee on Food Additives the previous provisional value of PTDI of inorganic arsenic ($2.1\text{ }\mu\text{g kg}^{-1}$ body wt. d^{-1}) was withdrawn (JEFCA, 2010). Instead of this value $3.0\text{ }\mu\text{g}$ per kg body wt. per day was recommended as the lower limit of inorganic arsenic for the benchmark dose for 0.5% ($\text{BMDL}_{0.5}$) increased frequency of lung cancer (FAO/WHO, 2010). On contrary, due to smaller value of $\text{BMDL}_{0.5}$ (lung cancer) relative to skin lesions ($5.4\text{ }\mu\text{g}$ per kg body wt. per day), carcinogenic risk should be given more attention compared to

non-carcinogenic health risk. However, Codex committee on contaminants and food have denied to agree the lower limit of $\text{BMDL}_{0.5}$ (Codex, 2011). Conversely, EFSA has made the decision for intake of $0.3\text{--}8\text{ }\mu\text{g kg}^{-1}$ body wt. d^{-1} to characterize the risk of inorganic arsenic (EFSA, 2009).

Halder et al. (2014) determined the estimated dietary intake of inorganic arsenic with the range between 0.10 and $4.79\text{ }\mu\text{g kg}^{-1}$ body wt. d^{-1} from the consumption of cooked rice in West Bengal (India). The value of dietary intake of inorganic arsenic exceeded the PTDI value of $2.1\text{ }\mu\text{g kg}^{-1}$ body wt. d^{-1} for more than 10% of the total population. By considering the RfD of $0.3\text{ }\mu\text{g kg}^{-1}$ body wt. d^{-1} , their estimated value of HQ was in the range of 0.33–15.7 with the average of 3.85 signifying adverse potential non-carcinogenic health risk among the population. Similarly, using the old US EPA cancer slope factor of $1.5\text{ per mg kg}^{-1}$ body wt. d^{-1} , they reported the value of CR with the mean of 2.51 per 10,000 numbers of the person which substantially recognizes the lifetime risk of cancer to the population through arsenic based rice diet. As the reference level, US EPA has fixed the upper limit of CR of 1 per 10,000 from any given source of arsenic (Tsuji et al., 2007). In another recent study, extreme higher value of CR was reported with the average of 12.3 per 10,000 indicating critical carcinogenic risk from the consumption of locally grown rice in Hunan province of China with the variation of dietary intake from $0.63\text{ to }1.49\text{ }\mu\text{g kg}^{-1}$ body wt. d^{-1} (Ma et al., 2016) but, which did not exceed the lower limit of $\text{BMDL}_{0.5}$. Consequently, the HQ was in the range of 2.12–4.98 with the mean of 2.73. Recently, Islam et al. (2017) found extreme higher value of CR and HQ with 2.88 per 1000 and 6.40 respectively from Narayanganj district of Bangladesh. The estimated daily intake of inorganic arsenic found to be $1.92\text{ }\mu\text{g kg}^{-1}$ body wt. d^{-1} which were much higher than mean value of $0.82\text{ }\mu\text{g kg}^{-1}$ body wt. d^{-1} as previously reported by Ma et al. (2016) from China. Their study also showed that female populations were much more susceptible to cancer risk compared to male at any group of age from inorganic arsenic ingestion. The average cancer risk factor was found to be 7.9 per 100,000 and 10.4 per 100,000 for female and male population among the all age of groups respectively from consumption of Taiwanese rice with the estimated average daily intake of $0.079\text{--}0.104\text{ }\mu\text{g kg}^{-1}$ body wt. d^{-1} (Chen et al., 2016). In the earlier study, Meharg et al. (2009) modelled the internal cancer risk on rice consuming populations among the different countries by using slope factor of $3.67\text{ per mg kg}^{-1}$ body wt. d^{-1} . Consumption of rice of 100 g d^{-1} and body wt. of 60 kg were taken into consideration. Median excess cancer rate in the decreasing sequence were 7.0, 5.4, 5.0, 4.3 and 3.6 per 10,000 for ingestion of polished rice from China, US, Bangladesh, Italy and India. When country specific rice consumption was considered, Bangladesh contributed maximum with the value of 22.1 per 10,000 followed by 15.2 per 10,000 for China, 6.9 per 10,000 for India, 1.3 per 10,000 for US and 0.7 per 10,000 for Italy. In another recent investigation of risk assessment, Huang et al. (2015) determined target hazard quotient (THQ) from the ratio of estimated daily intake of inorganic arsenic to skin lesions (JEFCA, 2011) to assess the potential non carcinogenic health risk based on the variation of inorganic arsenic content in Chinese milled rice samples. Results showed that Chinese population did not fall under non carcinogenic health risk on dietary exposure because the mean of the THQ values ranged from 0.060 to 0.144, not exceeded 1 irrespective of gender and all age groups. On the other hand, in the similar study, Margin of Exposure (MOE) was introduced according to the ratio of $\text{BMDL}_{0.5}$ (lung cancer) to estimated dietary intake of inorganic arsenic to define carcinogenic risk. The values of MOE ranged from 3.86 to 8.54 at the mean falling below 100 under all age groups and gender of Chinese consumer showing the reflectance of carcinogenic health risk from the exposure of rice based inorganic arsenic. This is well consistent with another finding by Li et al. (2015) where mean MOE value was 7.3 for Chinese adult from inorganic arsenic exposure in milled rice samples. Based on the per-capita consumption of rice, Chinese rice consumers exhibited much higher carcinogenic risk with the MOE ranged from 7.8 to 10.3. While

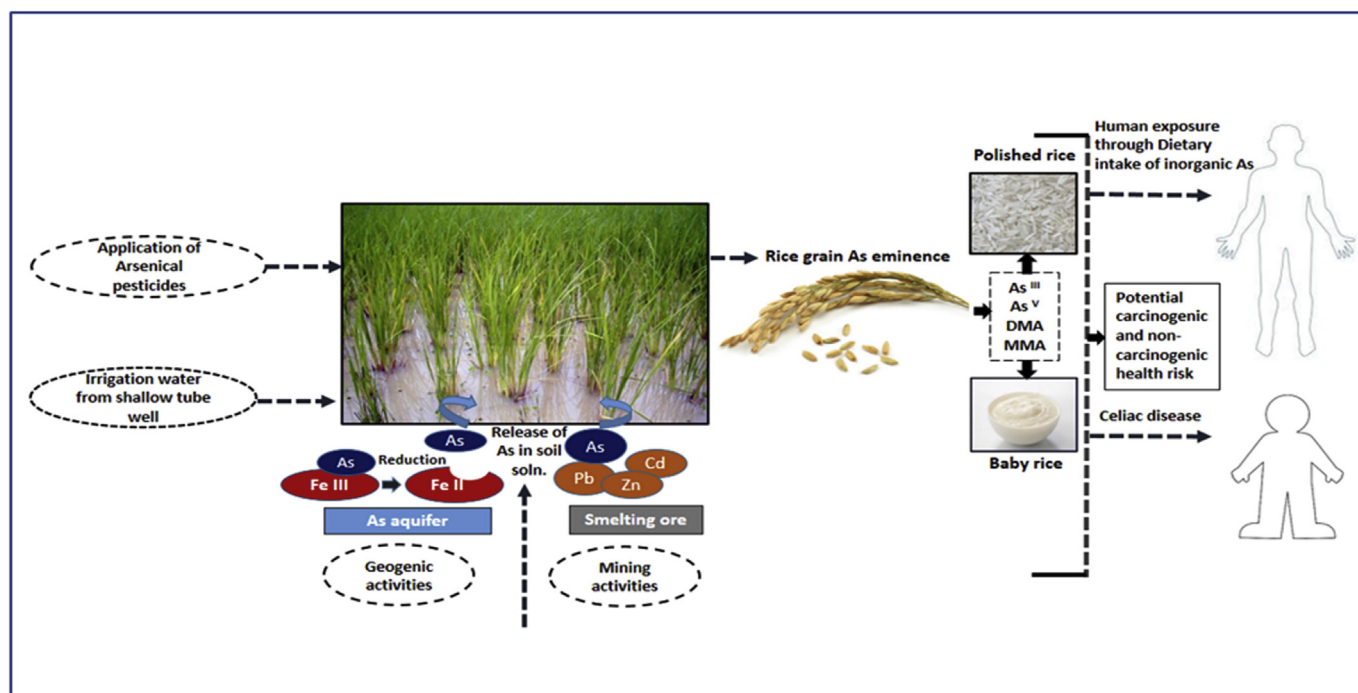


Fig. 3. Schematic representation of arsenic entering pathways in soil and its impact on rice and rice-based products in order to increase human health risk.

the consumptions of Indian and US rice were appeared to be less hazardous with the MOE of 30.9 and 54.6 respectively. Rice produced in Bangladesh and Thailand also contributed higher MOE compared to China. In risk assessment from the consumption of Indian rice, Kumar et al. (2016a) reported higher value of Health-Risk Index ($HRI > 1$) in gangetic plain of Bihar. Higher HRI values ranged from 13.6 to 219.5 were also reported from part of the Bihar (India) (Singh and Ghosh, 2012).

In the aspect on dietary contribution, Williams et al. (2005) found wide variation in contribution to PTDI ranged from 4 to 88% assuming body wt. of 60 kg with rice consumption rate of 0.5 kg d^{-1} based on the modelling of inorganic arsenic speciation of 15 different varieties from Bangladeshi market rice. The contribution to PTDI ranged from 8 to 50% by considering higher average American body wt. of 70 kg from US grown white and brown long grain variety followed by European (29–50%) and Indian origin (7–18%). In contrast to seasonal variation in Bangladesh, boro rice contributed higher proportion to PTDI as compared to aman rice with the ranged varied from 41 to 146% and 23–103% respectively (Williams et al., 2006). The higher contribution to PTDI under boro season was strongly associated with high grain arsenic level as well as utilisation of arsenic contaminated irrigation water from the consumption of Bangladeshi rice. In another recent study, Islam et al. (2017) showed that rice inorganic arsenic ingestion contributed to PTDI with the range of 16–81% from different districts of Bangladesh. Rice may contribute significantly to dietary exposure of average Americans if elevated rice grain inorganic arsenic is associated with the consumption of 1 L drinking water per day ($10 \mu\text{g L}^{-1}$ arsenic). Daily exposure of inorganic arsenic potentially exceeds the level of $10 \mu\text{g d}^{-1}$ (Williams et al., 2007a). Based on comparative assessment, Rahman et al. (2014) conducted a study to find out the risk contribution of Australian grown rice among consumer of Australian, Asian and European immigrants. Rice imported to Australia on sale from different countries were taken into consideration. Asian immigrants were estimated to contribute in maximum proportion to PTDI of inorganic arsenic in the range of 59.08–98.82% followed by average Australian (4.54–7.59%) and European immigrants (1.77–2.96%) from the consumption of Australian grown rice by considering the bioavailability of 98.3% and 0.33% for inorganic and organic arsenic

respectively. In contrast, among the imported products, Italian rice was served maximum on an average of 35.8% followed by Indian (12.4%) and Bangladeshi rice (9%) to PTDI. The rice consumption rates of 0.046, 0.016 and 0.44 kg d^{-1} were taken into account for Australian, European and Asian immigrants respectively.

7.2. Dietary intake of baby rice and celiac disease

The effect of arsenic concentration in baby rice is of great relevance to infants suffering celiac disease. As the intake of rice, baby rice, including other rice-based products like rice cereals, rice crackers may also contribute dietary exposure to inorganic arsenic to infants' diet. The rice-based products have been found to be associated with daily exposure to inorganic arsenic among the European children under 3 years of age (EFSA, 2009). However, health risk associated with the consumption of baby rice to infants, and young children has paid much attention in recent years. Celiac disease induced by a gluten based diet to children is of the major concern. According to statistical evidence, scientists have shown the prevalence of Celiac disease at 1 out of 118 and 1 out of 389 for children and adults in Spain respectively, whereas world prevalence has found at 1% (Munera-Picazo et al., 2014). Some of the recent studies postulated that 1% of children in US and Europe are affected by celiac disease through consumption of rice-based products (Da Sacco et al., 2013; Munera-Picazo et al., 2014; Newton and Singer, 2012). For instance, infants and young children possess much higher toxicity from inorganic arsenic because of their higher consumption rate of rice relative to body mass than adults (EFSA, 2009; Meharg and Zhao, 2012). However, many of the studies have focused on celiac disease from rice consistent diet, especially in part of the Europe and US (Munera-Picazo et al., 2014) investigated dietary intake of inorganic arsenic from different rice-based food collected from Spanish supermarket to predict health risk to celiac children in the age group of 1–5 year. From various kinds of food product fed to children, daily intake of inorganic arsenic appeared to be ranged from 0.60 to $0.84 \mu\text{g kg}^{-1}$ body wt. and 0.61 – $0.78 \mu\text{g kg}^{-1}$ body wt. for the children of age group 2–3 year and 3–5 year respectively. Carbonell-Barrachina et al. (2012) estimated daily intake of arsenic from Spanish infant products in different age groups. The mean daily intake of arsenic at

0.26, 0.27, 0.41, 0.40 $\mu\text{g kg}^{-1}$ body wt. d^{-1} was observed for celiac children with the consumption of gluten-free rice products in the age group of 4, 6, 8 and 12 months respectively. Comparatively lower arsenic intake was obtained as 0.05, 0.16, 0.25 and 0.26 $\mu\text{g kg}^{-1}$ body wt. d^{-1} in the same age group of children feeding cereals mixed with gluten. Higher inorganic arsenic bioavailability in gluten-free rice products was reflected the high risk to infants with celiac disease compared to cereals with gluten. On the other hand, the UK baby rice may be associated with higher risk of consumption. Meharg et al. (2008) postulated the median intake of 0.21 $\mu\text{g kg}^{-1}$ body wt. d^{-1} arsenic for one-year-old baby assuming the median body weight of 9.25 kg and daily consumption of 20 g portion from baby rice. According to US and EU regulation total and inorganic arsenic in drinking water were set up at 10 $\mu\text{g L}^{-1}$. Assuming the body wt. of 60 kg and daily consumption of 1 L drinking water the maximum intake of arsenic was predicted as 0.17 $\mu\text{g kg}^{-1}$ d^{-1} under EU/US recommendation (NRC, 2001). If a child takes 3 serving per day at 0.21 $\mu\text{g kg}^{-1}$, then the consumption would be 3.7 fold higher than US/EU regulation. Based on the investigation of Burló et al. (2012), daily intake of inorganic arsenic was estimated to be 0.16 $\mu\text{g kg}^{-1}$ d^{-1} for Spanish children with the age group from 4 to 6 months. This figure reached to 0.33 $\mu\text{g kg}^{-1}$ d^{-1} in the age group of 6–8 months, which was double the recommended threshold level of drinking water. The intake of age group from 8 to 12 months was similar to 2nd period. In another recent study, Signes-Pastor et al. (2016b) reported median intake of 0.7, 2.2 and 3.5 μg of arsenic per serving to infant from baby rice ($n = 29$), rice cereals ($n = 53$) and rice cracker ($n = 97$) of UK origin respectively. Whereas USFDA dataset represented the median intake of 1.7, 2.6 and 2.4 μg of arsenic per serving from baby rice ($n = 85$), rice cereals ($n = 105$) and rice crackers ($n = 199$) respectively (USFDA, 2013). Although, these all estimated values of dietary intake were below the maximum level of 8.0 $\mu\text{g kg}^{-1}$ body wt. d^{-1} proposed by EFSA (EFSA, 2009) but some of the values exceeded the lower limit of 0.3 $\mu\text{g kg}^{-1}$ body wt. d^{-1} for children with higher age group in particular. However, Munera-Picazo et al. (2014) proposed a recommendation on labelling of geographical origin as well as content of arsenic for manufactured rice-based products in order to concern about the celiac children in particular.

8. Concluding remarks

This review has attempted to find out the major components responsible for geographical variation of arsenic in the soil-rice system. Rice comprises its growth in the existence of anaerobic condition of paddy soil. Therefore, rice can accumulate higher levels of arsenic into grain among other cereal crops. Growing rice in the arsenic contaminated soil of different origin is believed to be a major route of arsenic exposure, which triggers a potential dietary risk to millions of people. Consequently, it critically impacts to sustainable rice production with adverse effect on overall food security. Increasing the level of inorganic arsenic concentration in rice grain seems to be a special health concern. A number of health hazards like internal cancer (liver, bladder, kidney and lung), skin cancer and diabetes may be associated with the elevated exposure to inorganic arsenic. However, the evidence of arsenic present in groundwater has found to be widespread throughout the world. Rice produced in the region with higher arsenic content thus causes health risk to the population.

Based on the geographical focus, the published articles contained in our meta-analysis estimation cover many important geographical areas of rice cultivation. The meta-analysis approach shows the impact of regional variability on rice grain arsenic availability. A number of investigations have revealed that irrigation of arsenic contaminated groundwater traditionally increases the rice grain arsenic concentration in parts of SE Asian countries, particularly for Bangladesh and India. The elevated level of inorganic arsenic in rice grain from these countries is associated with higher risk at the meta-analysis level with increasing the question of food safety. While environmental pollution through

mining activities may be probably associated to increase grain arsenic accumulation of rice grown in China and part of the European origin. In contrast, the source of arsenic in US grown rice is predominantly from application of arsenical pesticide. Although based on the meta-analysis estimation it has been clearly established that majority of the US rice was found to be the organic type than Asian and European origin.

Consumption of infant rice and rice-based products for the children suffering from celiac disease should be given special attention. Powdered baby rice and other related products like pasta, biscuits, and noodles are the major sources of inorganic arsenic ingestion to baby's diet. Country wise variation of inorganic arsenic concentration in baby rice and rice-based products may cause serious health risk for celiac infants. Elevated arsenic content in infant rice from UK, Spanish origin is fundamentally throwing a remarkable question, whether rice should be used as the ingredient for infant food formulation. In the question of high arsenic exposure, manufacturing companies should take the leading role on priority to impose the awareness from consumption of these particular types of foods. Details of information regarding the geographic origin, inorganic arsenic content should be ascertained with proper labelling.

Furthermore, we must explore and categorize the possible source of contamination across the rice-growing region which may critically enhance the bioavailability of arsenic in rice. This approach ultimately suggests to minimize the risk in the concern of future remedial activities.

Acknowledgement

We sincerely thanks to the Indian Statistical Institute for providing financial support for this study.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jenvman.2018.12.034>.

References

- Abedin, M.J., Cresser, M.S., Meharg, A.A., Feldmann, J., Cotter-Howells, J., 2002. Arsenic accumulation and metabolism in rice (*Oryza sativa* L.). *Environ. Sci. Technol.* 36, 962–968.
- Adomako, E.E., Solaiman, A., Williams, P.N., Deacon, C., Rahman, G., Meharg, A.A., 2009. Enhanced transfer of arsenic to grain for Bangladesh grown rice compared to US and EU. *Environ. Int.* 35, 476–479.
- Ahmed, K.M., Bhattacharya, P., Hasan, M.A., Akhter, S.H., Alam, S.M., Bhuyian, M.H., Imam, M.B., Khan, A.A., Sracek, O., 2004. Arsenic enrichment in groundwater of the alluvial aquifers in Bangladesh: an overview. *Appl. Geochem.* 19, 181–200.
- Ahsan, D.A., DelValls, T.A., Blasco, J., 2009. Distribution of arsenic and trace metals in the floodplain agricultural soil of Bangladesh. *Bull. Environ. Contam. Toxicol.* 82, 11–15.
- Álvarez-Ayuso, E., García-Sánchez, A., Querol, X., Moyano, A., 2008. Trace element mobility in soils seven years after the Aznalcóllar mine spill. *Chemosphere* 73, 1240–1246.
- Basu, A., Schreiber, M.E., 2013. Arsenic release from arsenopyrite weathering: insights from sequential extraction and microscopic studies. *J. Hazard. Mater.* 262, 896–904.
- Bednar, A., Garbarino, J., Ranville, J., Wildeman, T., 2002. Presence of organoarsenicals used in cotton production in agricultural water and soil of the southern United States. *J. Agric. Food Chem.* 50, 7340–7344.
- Bhattacharya, P., Samal, A., Majumdar, J., Santra, S., 2010. Accumulation of arsenic and its distribution in rice plant (*Oryza sativa* L.) in Gangetic West Bengal, India. *Paddy Water Environ.* 8, 63–70.
- Bissen, M., Frimmel, F.H., 2003. Arsenic—a review. Part I: occurrence, toxicity, speciation, mobility. *Clean. - Soil, Air, Water* 31, 9–18.
- Brammer, H., Ravenscroft, P., 2009. Arsenic in groundwater: a threat to sustainable agriculture in South and South-east Asia. *Environ. Int.* 35, 647–654.
- Burló, F., Ramírez-Gandolfo, A., Signes-Pastor, A.J., Haris, P.I., Carbonell-Barrachina, Á.A., 2012. Arsenic contents in Spanish infant rice, pureed infant foods, and rice. *J. Food Sci.* 77, T15–T19.
- Carbonell-Barrachina, Á.A., Wu, X., Ramírez-Gandolfo, A., Norton, G.J., Burló, F., Deacon, C., Meharg, A.A., 2012. Inorganic arsenic contents in rice-based infant foods from Spain, UK, China and USA. *Environ. Pollut.* 163, 77–83.
- Carey, A.M., Scheckel, K.G., Lombi, E., Newville, M., Choi, Y., Norton, G.J., Charnock, J.M., Feldmann, J., Price, A.H., Meharg, A.A., 2010. Grain unloading of arsenic species in rice. *Plant Physiol.* 152, 309–319.

- Chen, H.-L., Lee, C.-C., Huang, W.-J., Huang, H.-T., Wu, Y.-C., Hsu, Y.-C., Kao, Y.-T., 2016. Arsenic speciation in rice and risk assessment of inorganic arsenic in Taiwan population. *Environ. Sci. Pollut. Res.* 23, 4481–4488.
- Codex, 2011. Discussion Paper on Guidance for Risk Management Options on How to Deal with the Results from New Risk Assessment Methodologies (CX/CF 11/5/11). The Hague, the Netherlands: Codex Committee on Contaminants in Foods http://ftp.fao.org/codex/meetings/CCCF/CCCF5/cf05_11e.pdf.
- Cubadda, F., Ciardullo, S., D'Amato, M., Raggi, A., Aureli, F., Carcea, M., 2010. Arsenic contamination of the environment – food chain: a survey on wheat as a test plant to investigate phyto available arsenic in Italian agricultural soils and as a source of inorganic arsenic in the diet. *J. Agric. Food Chem.* 58, 10176–10183.
- D'ilio, S., Alessandrelli, M., Cresti, R., Forte, G., Caroli, S., 2002. Arsenic content of various types of rice as determined by plasma-based techniques. *Microchem. J.* 73, 195–201.
- Da Sacco, L., Baldassarre, A., Masotti, A., 2013. Diet's role in the toxicity of inorganic arsenic (iAs): a journey from soil to children's mouth. *J. Geochem. Explor.* 131, 45–51.
- De Miguel, E., Iribarren, I., Chacon, E., Ordóñez, A., Charlesworth, S., 2007. Risk-based evaluation of the exposure of children to trace elements in playgrounds in Madrid (Spain). *Chemosphere* 66, 505–513.
- Díez, M., Simón, M., Dorronsoro, C., García, I., Martín, F., 2007. Background arsenic concentrations in Southeastern Spanish soils. *Sci. Total Environ.* 378, 5–12.
- Dittmar, J., Voegelin, A., Roberts, L.C., Hug, S.J., Saha, G.C., Ali, M.A., Badruzzaman, A.B., Kretschmar, R., 2007. Spatial distribution and temporal variability of arsenic in irrigated rice fields in Bangladesh. 2. Paddy soil. *Environ. Sci. Technol.* 41, 5967–5972.
- Dittmar, J., Voegelin, A., Roberts, L.C., Hug, S.J., Saha, G.C., Ali, M.A., Badruzzaman, A.B.M., Kretschmar, R., 2010. Arsenic accumulation in a paddy field in Bangladesh: seasonal dynamics and trends over a three-year monitoring period. *Environ. Sci. Technol.* 44, 2925–2931.
- Dixit, S., Hering, J.G., 2003. Comparison of arsenic(V) and arsenic(III) sorption onto iron oxide minerals: implications for arsenic mobility. *Environ. Sci. Technol.* 37, 4182–4189.
- Dousova, B., Buzek, F., Rothwell, J., Krejčova, S., Lhotka, M., 2012. Adsorption behavior of arsenic relating to different natural solids: soils, stream sediments and peats. *Sci. Total Environ.* 433, 456–461.
- Duxbury, J.M., 2007. Remediation of Arsenic for Agriculture Sustainability, Food Security and Health in Bangladesh.
- Duxbury, J.M., Mayer, A.B., Lauren, J.G., Hassan, N., 2003. Food chain aspects of arsenic contamination in Bangladesh: effects on quality and productivity of rice. *J. Environ. Sci. Health A Tox. Hazard. Subst. Environ. Eng.* 38, 61–69.
- Dzombak, D.A., 1990. Surface Complexation Modeling: Hydrous Ferric Oxide. John Wiley & Sons.
- EFSA (European Food Safety Authority) Panel on Contaminants in the Food Chain, 2009. Scientific opinion on arsenic in food. EFSA J. 7 (10), 1351. <http://www.efsa.europa.eu/en/efsajournal/doc/1351.pdf>, Accessed date: 20 July 2014.
- FAO, 2017. Rice Market Monitor. http://www.fao.org/fileadmin/templates/est/COMM_MARKETS_MONITORING/Rice/Images/RMM/RMM-Jul17.pdf, Accessed date: July 2017.
- FAO/WHO, 2010. In: Summary and Conclusions of 72nd Meeting. JECFA, Rome.
- Farooqi, A., Masuda, H., Firdous, N., 2007. Toxic fluoride and arsenic contaminated groundwater in the Lahore and Kasur districts, Punjab, Pakistan and possible contaminant sources. *Environ. Pollut.* 145, 839–849.
- Feng, M., Schrlau, J.E., Snyder, R., Snyder, G.H., Chen, M., Cisar, J.L., Cai, Y., 2005. Arsenic transport and transformation associated with MSMA application on a golf course green. *J. Agric. Food Chem.* 53, 3556–3562.
- Fitz, W.J., Wenzel, W.W., 2002. Arsenic transformations in the soil-rhizosphere-plant system: fundamentals and potential application to phytoremediation. *J. Biotechnol.* 99, 259–278.
- García-Sánchez, A., Álvarez-Ayuso, E., 2003. Arsenic in soils and waters and its relation to geology and mining activities (Salamanca Province, Spain). *J. Geochem. Explor.* 80, 69–79.
- García-Sánchez, A., Moyano, A., Mayorga, P., 2005. High arsenic contents in groundwater of central Spain. *Environ. Geol.* 47, 847–854.
- García-Sánchez, A., Alonso-Rojó, P., Santos-Francés, F., 2010. Distribution and mobility of arsenic in soils of a mining area (Western Spain). *Sci. Total Environ.* 408, 4194–4201.
- Goldberg, S., 2002. Competitive adsorption of arsenate and arsenite on oxides and clay minerals. *Soil Sci. Soc. Am. J.* 66, 413–421.
- Goldberg, S., Johnston, C.T., 2001. Mechanisms of arsenic adsorption on amorphous oxides evaluated using macroscopic measurements, vibrational spectroscopy, and surface complexation modeling. *J. Colloid Interface Sci.* 234, 204–216.
- Halder, D., Biswas, A., Šlejko, Z., Chatterjee, D., Nriagu, J., Jacks, G., Bhattacharya, P., 2014. Arsenic species in raw and cooked rice: implications for human health in rural Bengal. *Sci. Total Environ.* 497, 200–208.
- Hamon, R.E., Lombi, E., Fortunati, P., Nolan, A.L., McLaughlin, M.J., 2004. Coupling speciation and isotope dilution techniques to study arsenic mobilization in the environment. *Environ. Sci. Technol.* 38, 1794–1798.
- Haque, S., Johannesson, K.H., 2006. Arsenic concentrations and speciation along a groundwater flow path: the Carrizo Sand aquifer, Texas, USA. *Chem. Geol.* 228, 57–71.
- Heitkemper, D.T., Vela, N.P., Stewart, K.R., Westphal, C.S., 2001. Determination of total and speciated arsenic in rice by ion chromatography and inductively coupled plasma mass spectrometry. *J. Anal. At. Spectrom.* 16, 299–306.
- Herbel, M., Fendorf, S., 2006. Biogeochemical processes controlling the speciation and transport of arsenic within iron coated sands. *Chem. Geol.* 228, 16–32.
- Hirata, S., Toshimitsu, H., Aihara, M., 2006. Determination of arsenic species in marine samples by HPLC-ICP-MS. *Anal. Sci.* 22, 39–43.
- Hite, A.H., 2013. Arsenic and rice: a call for regulation. *Nutrition* 29, 353.
- Huang, Y., Wang, M., Mao, X., Qian, Y., Chen, T., Zhang, Y., 2015. Concentrations of inorganic arsenic in milled rice from China and associated dietary exposure assessment. *J. Agric. Food Chem.* 63, 10838–10845.
- IARC (International Agency for Cancer Research), 2004. Some Drinking-water Disinfectants and Contaminants, Including Arsenic, vol. 84 IARC, Geneva, Switzerland.
- Islam, F.S., Gault, A.G., Boothman, C., Polya, D.A., Charnock, J.M., Chatterjee, D., Lloyd, J.R., 2004. Role of metal-reducing bacteria in arsenic release from Bengal delta sediments. *Nature* 430, 68–71.
- Islam, S., Rahman, M.M., Islam, M., Naidu, R., 2016. Arsenic accumulation in rice: consequences of rice genotypes and management practices to reduce human health risk. *Environ. Int.* 96, 139–155.
- Islam, S., Rahman, M.M., Islam, M., Naidu, R., 2017. Geographical variation and age-related dietary exposure to arsenic in rice from Bangladesh. *Sci. Total Environ.* 601, 122–131.
- Italian Grain, Feed Report, 2012. Maximum Levels of Contamination in Foods. Grain Report IT1212.
- Jahiruddin, M., Islam, M., Islam, M., Islam, S., 2004. Effects of as contamination on rice crop (*Oryza sativa* L.). *Environ. Tropica* 1, 104–110.
- Jain, A., Loeppert, R.H., 2000. Effect of competing anions on the adsorption of arsenate and arsenite by ferrihydrite. *J. Environ. Qual.* 29, 1422–1430.
- JECFA, 2011. Evaluation of Certain Contaminants in Food: Seventy-second Report of the Joint 310 FAO/WHO Expert Committee on Food Additives. WHO Technical Report Series. pp. 959.
- JECFA, 2012. Report of the Sixth Session of the Codex Committee on Contaminants in Foods. Joint FAO/WHO Food Standards Programme Codex Alimentarius Commission: Maastricht, The Netherlands, pp. 79.
- JECFA, 2010. Summary and Conclusions of 72nd Meeting. Joint FAO/WHO Expert Committee on Food Additives, Rome, Italy.
- Juhasz, A.L., Smith, E., Weber, J., Rees, M., Rofo, A., Kuchel, T., Sansom, L., Naidu, R., 2006. In vivo assessment of arsenic bioavailability in rice and its significance for human health risk assessment. *Environ. Health Perspect.* 114, 1826.
- Jung, M.C., Thornton, I., 1997. Environmental contamination and seasonal variation of metals in soils, plants and waters in the paddy fields around a Pb–Zn mine in Korea. *Sci. Total Environ.* 198, 105–121.
- Khan, M.A., Islam, M.R., Panaullah, G., Duxbury, J.M., Jahiruddin, M., Loeppert, R.H., 2009. Fate of irrigation-water arsenic in rice soils of Bangladesh. *Plant Soil* 322, 263–277.
- Khan, M.A., Islam, M.R., Panaullah, G., Duxbury, J.M., Jahiruddin, M., Loeppert, R.H., 2010a. Accumulation of arsenic in soil and rice under wetland condition in Bangladesh. *Plant Soil* 333, 263–274.
- Khan, M.A., Stroud, J.L., Zhu, Y.-G., McGrath, S.P., Zhao, F.-J., 2010b. Arsenic bioavailability to rice is elevated in Bangladeshi paddy soils. *Environ. Sci. Technol.* 44, 8515–8521.
- Kile, M.L., Houseman, E.A., Breton, C.V., Quamruzzaman, Q., Rahman, M., Mahiuddin, G., Christiani, D.C., 2007. Association between total ingested arsenic and toenail arsenic concentrations. *J. Environ. Sci. Health A Tox. Hazard. Subst. Environ. Eng.* 42, 1827–1834.
- Kumar, M., Rahman, M.M., Ramanathan, A., Naidu, R., 2016a. Arsenic and other elements in drinking water and dietary components from the middle Gangetic plain of Bihar, India: health risk index. *Sci. Total Environ.* 539, 125–134.
- Kumar, M., Ramanathan, A., Rahman, M.M., Naidu, R., 2016b. Concentrations of inorganic arsenic in groundwater, agricultural soils and subsurface sediments from the middle Gangetic plain of Bihar, India. *Sci. Total Environ.* 573, 1103–1114.
- Kwon, J.C., Nejad, Z.D., Jung, M.C., 2017. Arsenic and heavy metals in paddy soil and polished rice contaminated by mining activities in Korea. *Catena* 148, 92–100.
- Lafferty, B.J., Loeppert, R.H., 2005. Methyl arsenic adsorption and desorption behavior on iron oxides. *Environ. Sci. Technol.* 39, 2120–2127.
- Lamont, W.H., 2003. Concentration of inorganic arsenic in samples of white rice from the United States. *J. Food Compos. Anal.* 16, 687–695.
- Lee, S., 2006. Geochemistry and partitioning of trace metals in paddy soils affected by metal mine tailings in Korea. *Geoderma* 135, 26–37.
- Li, R., Stroud, J., Ma, J., McGrath, S., Zhao, F., 2009. Mitigation of arsenic accumulation in rice with water management and silicon fertilization. *Environ. Sci. Technol.* 43, 3778–3783.
- Li, J., Dong, F., Lu, Y., Yan, Q., Shim, H., 2014. Mechanisms controlling arsenic uptake in rice grown in mining impacted regions in South China. *PLoS One* 9, e108300.
- Li, X., Xie, K., Yue, B., Gong, Y., Shao, Y., Shang, X., Wu, Y., 2015. Inorganic arsenic contamination of rice from Chinese major rice-producing areas and exposure assessment in Chinese population. *Sci. China Chem.* 58, 1898–1905.
- Liao, X.-Y., Chen, T.-B., Xie, H., Liu, Y.-R., 2005. Soil as contamination and its risk assessment in areas near the industrial districts of Chenzhou City, Southern China. *Environ. Int.* 31, 791–798.
- Lin, K., Lu, S., Wang, J., Yang, Y., 2015a. The arsenic contamination of rice in Guangdong Province, the most economically dynamic provinces of China: arsenic speciation and its potential health risk. *Environ. Geochem. Health* 37, 353–361.
- Lin, S.-C., Chang, T.-K., Huang, W.-D., Lur, H.-S., Shyu, G.-S., 2015b. Accumulation of arsenic in rice plant: a study of an arsenic-contaminated site in Taiwan. *Paddy Water Environ.* 13, 11–18.
- Liu, H., Probst, A., Liao, B., 2005. Metal contamination of soils and crops affected by the Chenzhou lead/zinc mine spill (Hunan, China). *Sci. Total Environ.* 339, 153–166.
- Liu, W., Zhu, Y., Hu, Y., Williams, P., Gault, A., Meharg, A., Charnock, J., Smith, F., 2006. Arsenic sequestration in iron plaque, its accumulation and speciation in mature rice

- plants (*Oryza sativa* L.). *Environ. Sci. Technol.* 40, 5730–5736.
- Ljung, K., Palm, B., Grandér, M., Vahter, M., 2011. High concentrations of essential and toxic elements in infant formula and infant foods—A matter of concern. *Food Chem.* 127, 943–951.
- Loebenstein, J.R., 1994. Materials Flow of Arsenic in the United States. US Bureau of Mines.
- Lombi, E., Sletten, R., Wenzel, W., 2000. Sequentially extracted arsenic from different size fractions of contaminated soils. *Water Air Soil Pollut.* 124, 319–332.
- Ma, L., Wang, L., Jia, Y., Yang, Z., 2016. Arsenic speciation in locally grown rice grains from Hunan Province, China: Spatial distribution and potential health risk. *Sci. Total Environ.* 557, 438–444.
- Munera-Picazo, S., Ramírez-Gandolfo, A., Burló, F., Carbonell-Barrachina, Á.A., 2014. Inorganic and total arsenic contents in rice-based foods for children with celiac disease. *J. Food Sci.* 79, T122–T128.
- Mahimairaja, S., Bolan, N., Adriano, D., Robinson, B., 2005. Arsenic contamination and its risk management in complex environmental settings. *Adv. Agron.* 86, 1–82.
- Mandal, B.K., Suzuki, K.T., 2002. Arsenic round the world: a review. *Talanta* 58, 201–235.
- Manning, B.A., Goldberg, S., 1997. Adsorption and stability of arsenic (III) at the clay mineral – water interface. *Environ. Sci. Technol.* 31, 2005–2011.
- Manning, B.A., Fendorf, S.E., Goldberg, S., 1998. Surface structures and stability of arsenic (III) on goethite: spectroscopic evidence for inner-sphere complexes. *Environ. Sci. Technol.* 32, 2383–2388.
- Martin, M., Bonifacio, E., Hossain, K.J., Huq, S.I., Barberis, E., 2014. Arsenic fixation and mobilization in the soils of the Ganges and Meghna floodplains. Impact of pedo-environmental properties. *Geoderma* 228, 132–141.
- Martin, M., Stanchi, S., Hossain, K.J., Huq, S.I., Barberis, E., 2015. Potential phosphorus and arsenic mobilization from Bangladesh soils by particle dispersion. *Sci. Total Environ.* 536, 973–980.
- Masscheleyn, P.H., Delaune, R.D., Patrick Jr., W.H., 1991. Effect of redox potential and pH on arsenic speciation and solubility in a contaminated soil. *Environ. Sci. Technol.* 25, 1414–1419.
- Meharg, A.A., Rahman, M.M., 2003. Arsenic contamination of Bangladesh paddy field soils: implications for rice contribution to arsenic consumption. *Environ. Sci. Technol.* 37, 229–234.
- Meharg, A.A., Zhao, F.-J., 2012. *Arsenic & Rice*. Springer Science & Business Media.
- Meharg, A.A., Sun, G., Williams, P.N., Adomako, E., Deacon, C., Zhu, Y.G., Feldmann, J., Raab, A., 2008. Inorganic arsenic levels in baby rice are of concern. *Environ. Pollut.* 152, 746–749.
- Meharg, A.A., Williams, P.N., Adomako, E., Lawgali, Y.Y., Deacon, C., Villada, A., Cambell, R.C., Sun, G., Zhu, Y.G., Feldmann, J., Raab, A., Zhao, F.J., Islam, R., Hossain, S., Yanai, J., 2009. Geographical variation in total and inorganic arsenic content of polished (white) rice. *Environ. Sci. Technol.* 43, 1612–1617.
- Meliker, J.R., Franzblau, A., Slotnick, M.J., Nriagu, J.O., 2006. Major contributors to inorganic arsenic intake in southeastern Michigan. *Int. J. Hyg. Environ. Health* 209, 399–411.
- Moyano, A., Garcia-Sanchez, A., Mayorga, P., Anawar, H., Alvarez-Ayuso, E., 2009. Impact of irrigation with arsenic-rich groundwater on soils and crops. *J. Environ. Monit.* 11, 498–502.
- Munera-Picazo, S., Burló, F., Carbonell-Barrachina, Á.A., 2014. Arsenic speciation in rice-based food for adults with celiac disease. *Food Addit. Contam. Part A* 31, 1358–1366.
- Neumann, R.B., St. Vincent, A.P., Roberts, L.C., Badruzzaman, A.B.M., Ali, M.A., Harvey, C.F., 2011. Rice field geochemistry and hydrology: an explanation for why groundwater irrigated fields in Bangladesh are net sinks of arsenic from groundwater. *Environ. Sci. Technol.* 45, 2072–2078.
- Newton, K.P., Singer, S.A., 2012. Celiac Disease in Children and Adolescents: Special Considerations, Seminars in Immunopathology. Springer, pp. 479–496.
- Nordstrom, D.K., 2002. Worldwide Occurrences of Arsenic in Ground Water. American Association for the Advancement of Science.
- Norton, G.J., Duan, G., Dasgupta, T., Islam, M.R., Lei, M., Zhu, Y., Deacon, C.M., Moran, A.C., Islam, S., Zhao, F.J., Stroud, J.L., McGrath, S.P., Feldmann, J., Price, A.H., Meharg, A.A., 2009. Environmental and genetic control of arsenic accumulation and speciation in rice grain: comparing a range of common cultivars grown in contaminated sites across Bangladesh, China, and India. *Environ. Sci. Technol.* 43, 8381–8386.
- NRC (National Research Council), 2001. Arsenic in Drinking Water D 2001 Update. National Academy Press, Washington, DC.
- Ohno, K., Yanase, T., Matsuo, Y., Kimura, T., Rahman, M.H., Magara, Y., Matsui, Y., 2007. Arsenic intake via water and food by a population living in an arsenic-affected area of Bangladesh. *Sci. Total Environ.* 381, 68–76.
- Ori, L., Amacher, M., Sedberry Jr., J., 1993. Survey of the total arsenic content in soils in Louisiana. *Commun. Soil Sci. Plant Anal.* 24, 2321–2332.
- Patel, K., Shrivastava, K., Brandt, R., Jakubowski, N., Corns, W., Hoffmann, P., 2005. Arsenic contamination in water, soil, sediment and rice of central India. *Environ. Geochem. Health* 27, 131–145.
- Pigna, M., Caporale, A.G., Cavalca, L., Sommella, A., Violante, A., 2015. Arsenic in the soil environment: mobility and phytoavailability. *Environ. Eng. Sci.* 32, 551–563.
- Rahman, M.A., Hasegawa, H., Rahman, M.A., Rahman, M.M., Miah, M.M., 2006. Influence of cooking method on arsenic retention in cooked rice related to dietary exposure. *Sci. Total Environ.* 370, 51–60.
- Rahman, M.A., Hasegawa, H., Rahman, M.M., Rahman, M.A., Miah, M., 2007. Accumulation of arsenic in tissues of rice plant (*Oryza sativa* L.) and its distribution in fractions of rice grain. *Chemosphere* 69, 942–948.
- Rahman, M.M., Owens, G., Naidu, R., 2009. Arsenic levels in rice grain and assessment of daily dietary intake of arsenic from rice in arsenic-contaminated regions of Bangladesh—implications to groundwater irrigation. *Environ. Geochem. Health* 31, 179–187.
- Rahman, M.A., Rahman, M.M., Reichman, S.M., Lim, R.P., Naidu, R., 2014. Arsenic speciation in Australian-grown and imported rice on sale in Australia: implications for human health risk. *J. Agric. Food Chem.* 62, 6016–6024.
- Ramos-Miras, J., Díaz-Fernández, P., SanJosé-Very, A., Rodríguez-Martin, J., Roca, N., Bech, J., Roca-Perez, L., Boluda, R., Gil, C., 2014. Influence of parent material and soil use on arsenic forms in soils: a case study in the Amblés Valley (Castilla-León, Spain). *J. Geochem. Explor.* 147, 260–267.
- Reed, J., Sturgis, M., 1936. 1726001. Toxicity from arsenic compounds to rice on flooded soils. *Agron. J.* 28, 432–436.
- Roberts, L.C., Hug, S.J., Dittmar, J., Voegelin, A., Kretzschmar, R., Wehrli, B., Cirpka, O.A., Saha, G.C., Ali, M.A., Badruzzaman, A.B.M., 2010. Arsenic release from paddy soils during monsoon flooding. *Nat. Geosci.* 3, 53.
- Roychowdhury, T., 2008. Impact of sedimentary arsenic through irrigated groundwater on soil, plant, crops and human continuum from Bengal delta: special reference to raw and cooked rice. *Food Chem. Toxicol. Int. J. Publ. Br. Ind. Biol. Res. Assoc.* 46, 2856–2864.
- Roychowdhury, T., Uchino, T., Tokunaga, H., Ando, M., 2002. Survey of arsenic in food composites from an arsenic-affected area of West Bengal, India. *Food Chem. Toxicol. Int. J. Publ. Br. Ind. Biol. Res. Assoc.* 40, 1611–1621.
- Roychowdhury, T., Tokunaga, H., Ando, M., 2003. Survey of arsenic and other heavy metals in food composites and drinking water and estimation of dietary intake by the villagers from an arsenic-affected area of West Bengal, India. *Sci. Total Environ.* 308, 15–35.
- Sadiq, M., 1997. Arsenic chemistry in soils: an overview of thermodynamic predictions and field observations. *Water Air Soil Pollut.* 93, 117–136.
- Saha, G.C., Ali, M.A., 2007. Dynamics of arsenic in agricultural soils irrigated with arsenic contaminated groundwater in Bangladesh. *Sci. Total Environ.* 379, 180–189.
- Sanyal, S., Nasar, S., 2002. Arsenic contamination of groundwater in West Bengal (India): build-up in soil-crop systems. In: International Conference on Water Related Disasters. Kolkata. 5th–6th December.
- Sanz, E., Munoz-Olivas, R., Camara, C., Sengupta, M.K., Ahamed, S., 2007. Arsenic speciation in rice, straw, soil, hair and nails samples from the arsenic-affected areas of Middle and Lower Ganga plain. *J. Environ. Sci. Health Part A* 42, 1695–1705.
- Sarkar, D., Datta, R., Sharma, S., 2005. Fate and bioavailability of arsenic in organo-arsenical pesticide-applied soils: Part-I: incubation study. *Chemosphere* 60, 188–195.
- Sarkar, S., Basu, B., Kundu, C., Patra, P., 2012. Deficit irrigation: an option to mitigate arsenic load of rice grain in West Bengal, India. *Agric. Ecosyst. Environ.* 146, 147–152.
- Schoof, R., Yost, L., Eickhoff, J., Creclius, E., Cragin, D., Meacher, D., Menzel, D., 1999. A market basket survey of inorganic arsenic in food. *Food Chem. Toxicol.* 37, 839–846.
- Sherman, D.M., Randall, S.R., 2003. Surface complexation of arsenic (V) to iron (III) (hydr) oxides: structural mechanism from ab initio molecular geometries and EXAFS spectroscopy. *Geochem. Cosmochim. Acta* 67, 4223–4230.
- Shrivastava, A., Barla, A., Singh, S., Mandraha, S., Bose, S., 2017. Arsenic contamination in agricultural soils of Bengal deltaic region of West Bengal and its higher assimilation in monsoon rice. *J. Hazard. Mater.* 324, 526–534.
- Signes-Pastor, A.J., Carey, M., Carbonell-Barrachina, A.A., Moreno-Jiménez, E., Green, A.J., Meharg, A.A., 2016a. Geographical variation in inorganic arsenic in paddy field samples and commercial rice from the Iberian Peninsula. *Food Chem.* 202, 356–363.
- Signes-Pastor, A.J., Carey, M., Meharg, A.A., 2016b. Inorganic arsenic in rice-based products for infants and young children. *Food Chem.* 191, 128–134.
- Singh, S.K., Ghosh, A.K., 2012. Health risk assessment due to groundwater arsenic contamination: children are at high risk. *Hum. Ecol. Risk Assess.* Int. J. 18, 751–766.
- Smedley, P., Kinniburgh, D., 2002. A review of the source, behaviour and distribution of arsenic in natural waters. *Appl. Geochem.* 17, 517–568.
- Smith, P.G., 2007. Arsenic Biotransformations in Terrestrial Organisms: a Study of the Transport and Transformation of Arsenic in Plants, Fungi, Fur and Feathers, Using Conventional Speciation Analysis and X-ray Absorption Spectroscopy.
- Smith, E., Naidu, R., Alston, A., 1999. Chemistry of arsenic in soils: I. Sorption of arsenate and arsenite by four Australian soils. *J. Environ. Qual.* 28, 1719–1726.
- Sommella, A., Deacon, C., Norton, G., Pigna, M., Violante, A., Meharg, A., 2013. Total arsenic, inorganic arsenic, and other elements concentrations in Italian rice grain varies with origin and type. *Environ. Pollut.* 181, 38–43.
- Srivastava, S., Sharma, Y.K., 2013. Arsenic occurrence and accumulation in soil and water of eastern districts of Uttar Pradesh, India. *Environ. Monit. Assess.* 185, 4995–5002.
- Stachowicz, M., Hiemstra, T., van Riemsdijk, W.H., 2006. Surface speciation of as (III) and as (V) in relation to charge distribution. *J. Colloid Interface Sci.* 302, 62–75.
- Stroud, J.L., Norton, G.J., Islam, M.R., Dasgupta, T., White, R.P., Price, A.H., Meharg, A.A., McGrath, S.P., Zhao, F.J., 2011. The dynamics of arsenic in four paddy fields in the Bengal delta. *Environ. Pollut.* 159, 947–953.
- Su, Y.-H., McGrath, S.P., Zhao, F.-J., 2010. Rice is more efficient in arsenite uptake and translocation than wheat and barley. *Plant Soil* 328, 27–34.
- Sun, G.X., Williams, P.N., Carey, A.M., Zhu, Y.G., Deacon, C., Raab, A., Feldmann, J., Islam, R.M., Meharg, A.A., 2008. Inorganic arsenic in rice bran and its products are an order of magnitude higher than in bulk grain. *Environ. Sci. Technol.* 42, 7542–7546.
- Sun, G.-X., Williams, P.N., Zhu, Y.-G., Deacon, C., Carey, A.-M., Raab, A., Feldmann, J., Meharg, A.A., 2009. Survey of arsenic and its speciation in rice products such as breakfast cereals, rice crackers and Japanese rice condiments. *Environ. Int.* 35, 473–475.
- Takahashi, Y., Minamikawa, R., Hattori, K.H., Kurishima, K., Kihou, N., Yuita, K., 2004. Arsenic behavior in paddy fields during the cycle of flooded and non-flooded periods. *Environ. Sci. Technol.* 38, 1038–1044.

- Tattibayeva, D., Nebot, C., Miranda, J., Cepeda, A., Mateyev, E., Erkebaev, M., Franco, C., 2016. A study on toxic and essential elements in rice from the Republic of Kazakhstan: comparing the level of contamination in rice from the European Community. *Environ. Geochem. Health* 38, 85–98.
- Tenni, D., Martin, M., Barberis, E., Beone, G.M., Miniotti, E., Sodano, M., Zanzo, E., Fontanella, M.C., Romani, M., 2017. Total as and as speciation in Italian rice as related to producing areas and paddy soils properties. *J. Agric. Food Chem.* 65, 3443–3452.
- Torres-Escribano, S., Leal, M., Vélez, D., Montoro, R., 2008. Total and inorganic arsenic concentrations in rice sold in Spain, effect of cooking, and risk assessments. *Environ. Sci. Technol.* 42, 3867–3872.
- Tsuji, J.S., Yost, L.J., Barraj, L.M., Scrafford, C.G., Mink, P.J., 2007. Use of background inorganic arsenic exposures to provide perspective on risk assessment results. *Regul. Toxicol. Pharmacol.* 48, 59–68.
- Tyler, G., Olsson, T., 2001. Concentrations of 60 elements in the soil solution as related to the soil acidity. *Eur. J. Soil Sci.* 52, 151–165.
- Ullah, S., 1998. Arsenic Contamination of Groundwater and Irrigated Soils of Bangladesh, International Conference on Arsenic Pollution of Groundwater in Bangladesh: Causes, Effects and Remedies. Dhaka Community Hospital, Dhaka.
- USFDA (US Food and Drug Administration), 2013. **Arsenic in Rice: Full Analytical Results from Rice/rice Products.** Available: <http://www.fda.gov/Food/FoodborneIllnessContaminants/Metals/ucm319916.htm>, Accessed date: 23 July 2014.
- Williams, P.N., Price, A.H., Raab, A., Hossain, S.A., Feldmann, J., Meharg, A.A., 2005. Variation in arsenic speciation and concentration in paddy rice related to dietary exposure. *Environ. Sci. Technol.* 39, 5531–5540.
- Williams, P.N., Islam, M.R., Adomako, E.E., Raab, A., Hossain, S.A., Zhu, Y.G., Feldmann, J., Meharg, A.A., 2006. Increase in rice grain arsenic for regions of Bangladesh irrigating paddies with elevated arsenic in groundwaters. *Environ. Sci. Technol.* 40, 4903–4908.
- Williams, P.N., Raab, A., Feldmann, J., Meharg, A.A., 2007a. Market basket survey shows elevated levels of as in South Central U.S. processed rice compared to California: consequences for human dietary exposure. *Environ. Sci. Technol.* 41, 2178–2183.
- Williams, P.N., Villada, A., Deacon, C., Raab, A., Figuerola, J., Green, A.J., Feldmann, J., Meharg, A.A., 2007b. Greatly enhanced arsenic shoot assimilation in rice leads to elevated grain levels compared to wheat and barley. *Environ. Sci. Technol.* 41, 6854–6859.
- Williams, P.N., Lei, M., Sun, G., Huang, Q., Lu, Y., Deacon, C., Meharg, A.A., Zhu, Y.-G., 2009. Occurrence and partitioning of cadmium, arsenic and lead in mine impacted paddy rice: Hunan, China. *Environ. Sci. Technol.* 43, 637–642.
- Williams, P.N., Zhang, H., Davison, W., Meharg, A.A., Hossain, M., Norton, G.J., Brammer, H., Islam, M.R., 2011. Organic matter-solid phase interactions are critical for predicting arsenic release and plant uptake in Bangladesh paddy soils. *Environ. Sci. Technol.* 45, 6080–6087.
- Winkel, L., Berg, M., Amini, M., Hug, S.J., Johnson, C.A., 2008. Predicting groundwater arsenic contamination in Southeast Asia from surface parameters. *Nat. Geosci.* 1, ngeo254.
- Xie, Z.M., Huang, C.Y., 1998. Control of arsenic toxicity in rice plants grown on an arsenic-polluted paddy soil. *Commun. Soil Sci. Plant Anal.* 29, 2471–2477.
- Xu, X., McGrath, S., Meharg, A., Zhao, F., 2008. Growing rice aerobically markedly decreases arsenic accumulation. *Environ. Sci. Technol.* 42, 5574–5579.
- Yamaguchi, N., Nakamura, T., Dong, D., Takahashi, Y., Amachi, S., Makino, T., 2011. Arsenic release from flooded paddy soils is influenced by speciation, Eh, pH, and iron dissolution. *Chemosphere* 83, 925–932.
- Yang, N., Winkel, L.H., Johannesson, K.H., 2014. Predicting geogenic arsenic contamination in shallow groundwater of South Louisiana, United States. *Environ. Sci. Technol.* 48, 5660–5666.
- Zavala, Y.J., Duxbury, J.M., 2008. Arsenic in rice: I. Estimating normal levels of total arsenic in rice grain. *Environ. Sci. Technol.* 42, 3856–3860.
- Zavala, Y.J., Gerads, R., Gorleyok, H., Duxbury, J.M., 2008. Arsenic in rice: II. Arsenic speciation in USA grain and implications for human health. *Environ. Sci. Technol.* 42, 3861–3866.
- Zhao, F.-J., McGrath, S.P., Meharg, A.A., 2010. Arsenic as a food chain contaminant: mechanisms of plant uptake and metabolism and mitigation strategies. *Annu. Rev. Plant Biol.* 61, 535–559.
- Zhao, F.J., Zhu, Y.G., Meharg, A.A., 2013. Methylated arsenic species in rice: geographical variation, origin, and uptake mechanisms. *Environ. Sci. Technol.* 47, 3957–3966.
- Zhu, Y.G., Sun, G.X., Lei, M., Teng, M., Liu, Y.X., Chen, N.C., Wang, L.H., Carey, A.M., Deacon, C., Raab, A., Meharg, A.A., Williams, P.N., 2008. High percentage inorganic arsenic content of mining impacted and nonimpacted Chinese rice. *Environ. Sci. Technol.* 42, 5008–5013.